Fundamentals of Circuits and Electronics

CSAI 205

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Energy Bands of Solids

1. Valence Band

There are valence electrons in this energy band. Electrons can occupy this band entirely or partially, but it can never be empty. This band's electrons are unable to obtain enough energy from an outside electric field to participate in current conduction.

2. Conduction Band

This band contains conductive electrons. This band has either no electrons or some electrons that are only partially present. **Electrons** in this band contribute to current conduction.

Energy band gap

Energy band gap (E_g) is the minimal energy needed to move electrons from the valence band to the conduction band. It is the space between the conduction band's bottom and the valence band's top.

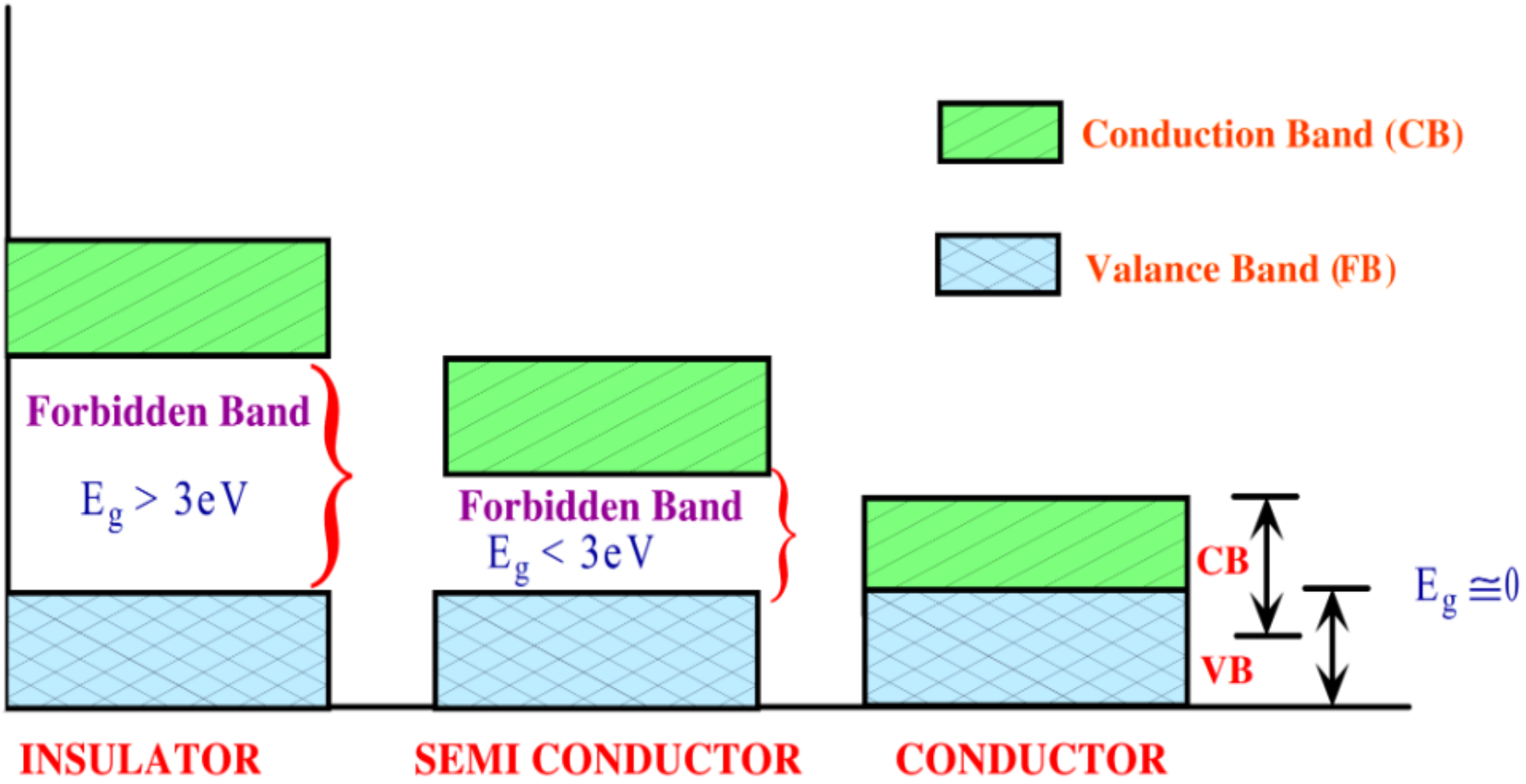
$\mathbf{E_g = hv = hc/\lambda}$

λ = wavelength

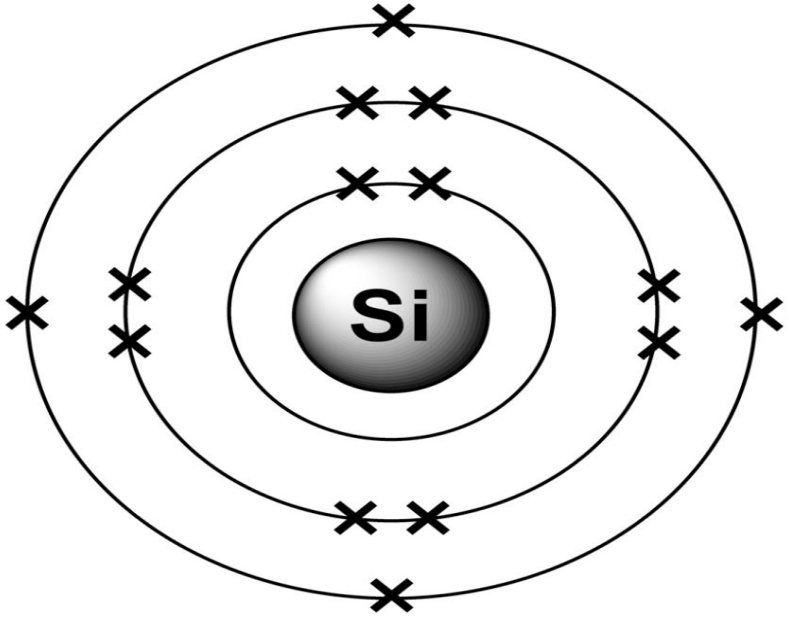
h = Planck's constant

c = velocity of light

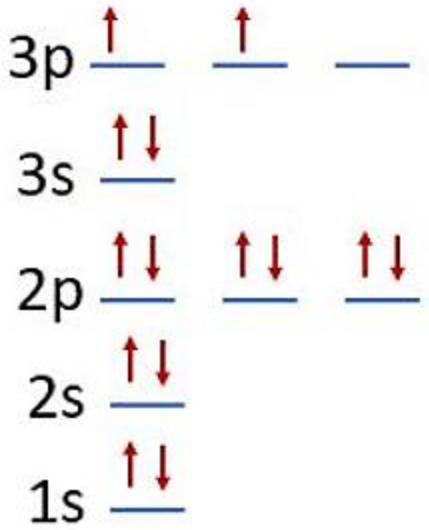
Conductors, Insulators and Semiconductors



Example for semiconductor: Silicon (14)



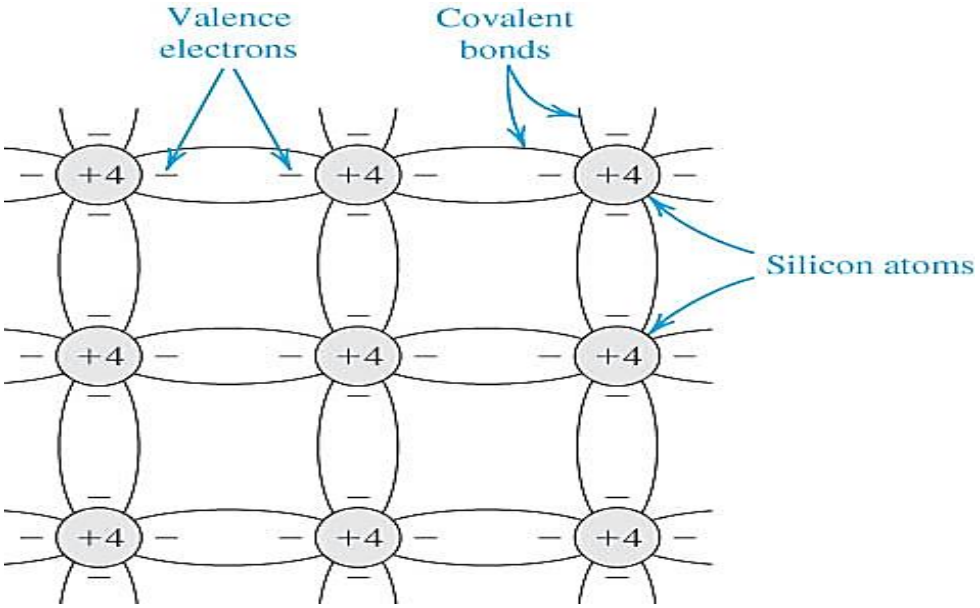
- **Bohr Model** (Early atomic model)
- semi-classical atomic model
- It obeys **($2n^2$)** model



- **Quantum Model (**Schrödinger model**)**
- It is called **quantum wave mechanics**
- **In our example,** The last subshell **(3p)** needs 4 more electrons to achieve stability

Intrinsic semiconductors

- Intrinsic semi conductor: is a material which has **not had any impurities** added intrinsic
- by sharing electrons with other atoms in covalent bonds, this sharing allows silicon atoms to participate in structures like silicon crystals, where each atom is effectively surrounded by a full set of 8 electrons

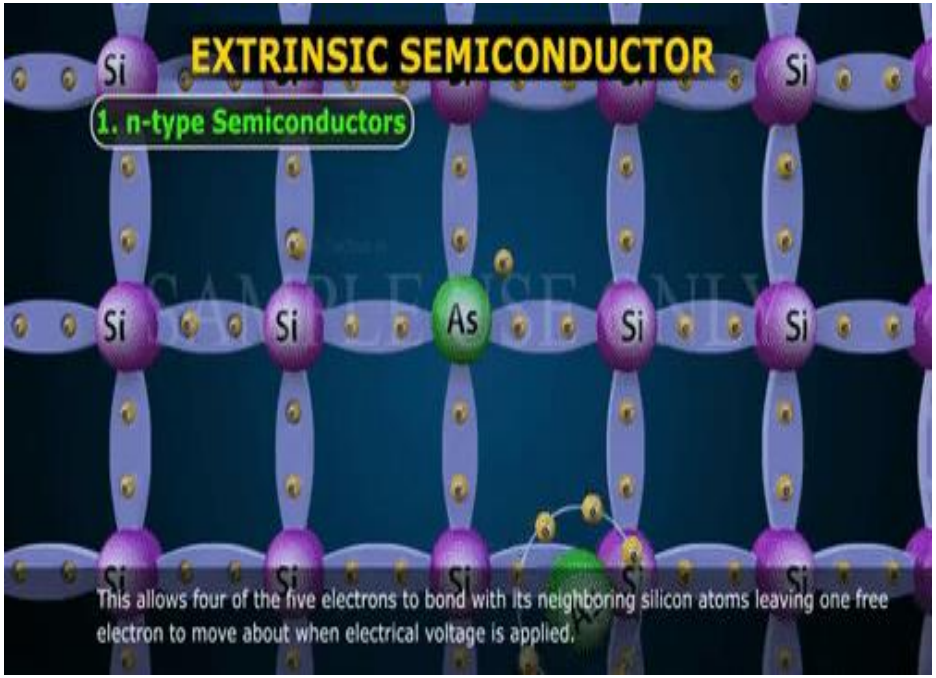


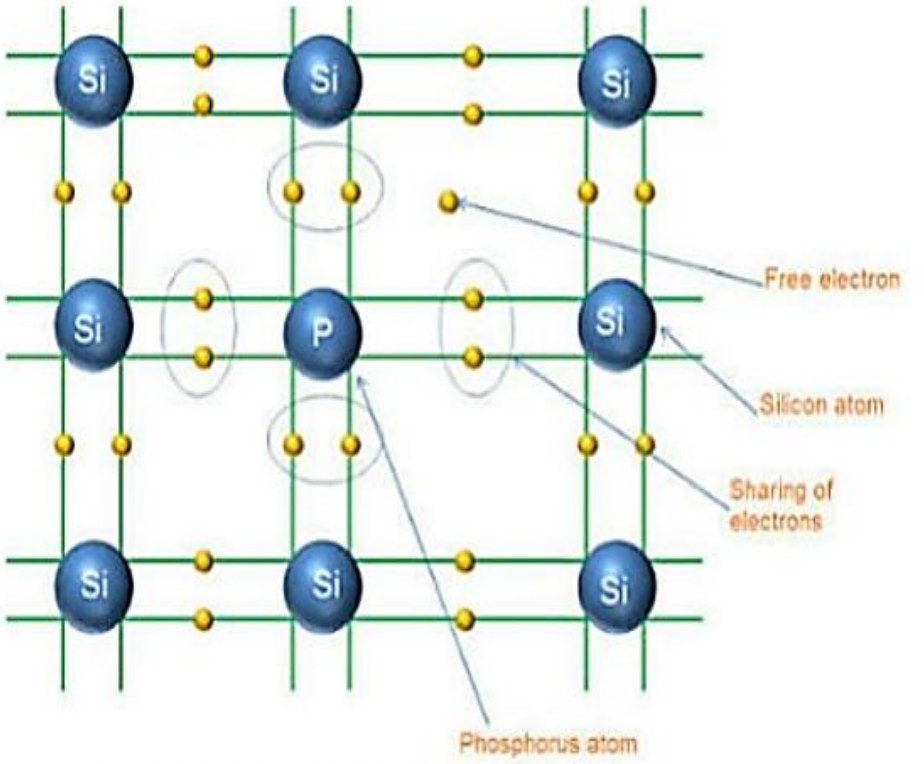
Extrinsic Semiconductors (doping process)

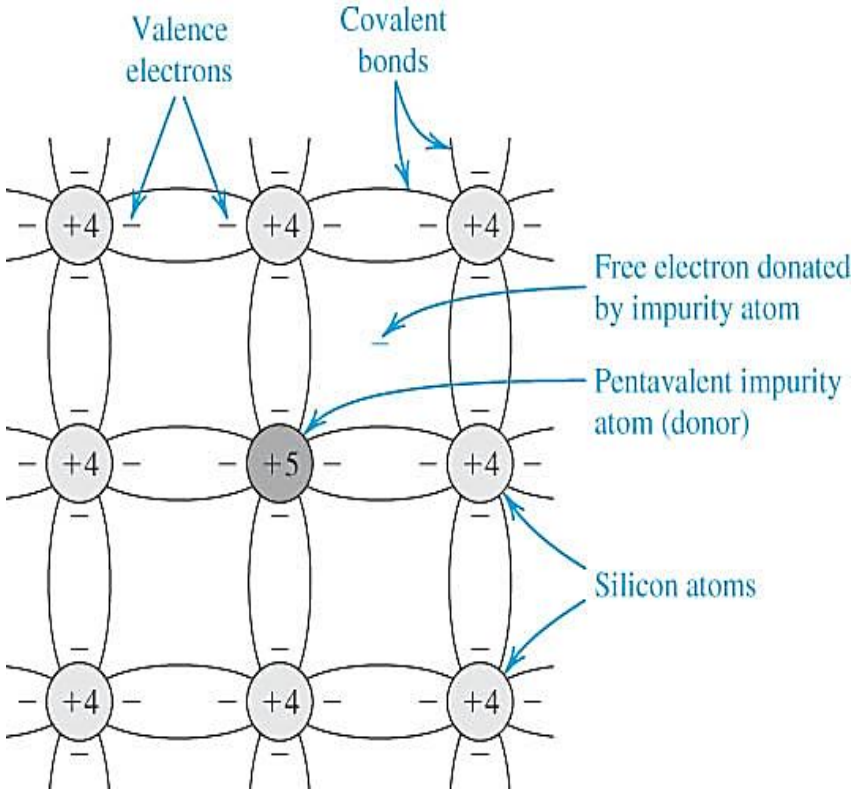
- Extrinsic semiconductor is a semiconductor crystal **doped with impurities**.
- We have 2 types: N-type semiconductors and P- type semiconductors

1. N- type semiconductors (Donors)

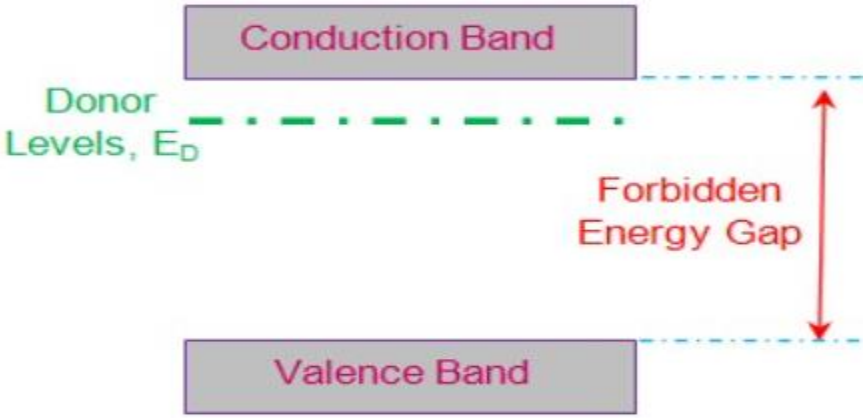
- To produce n-type semiconductor, silicon has to be doped with a **Pentavalent** impurity such as arsenic (As) and phosphorous (P)
- because the phosphorous atoms (for example) that is replace some of the silicon atoms in crystal structure have **five valence electrons**, four of them form bonds with neighboring silicon atoms while the **fifth** becomes a free electron

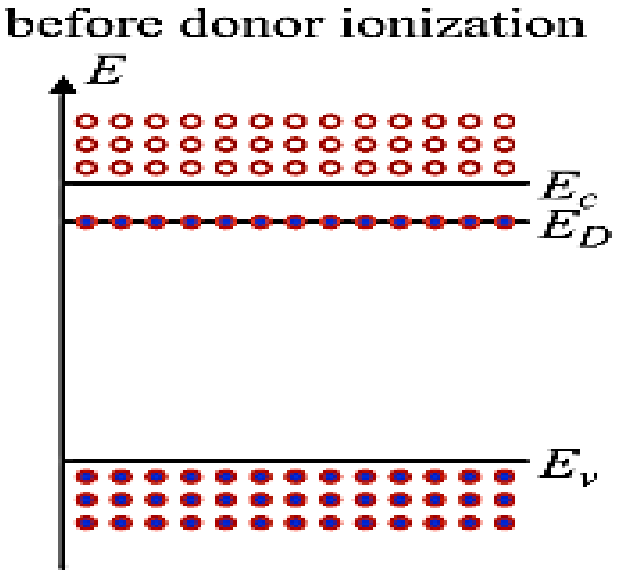


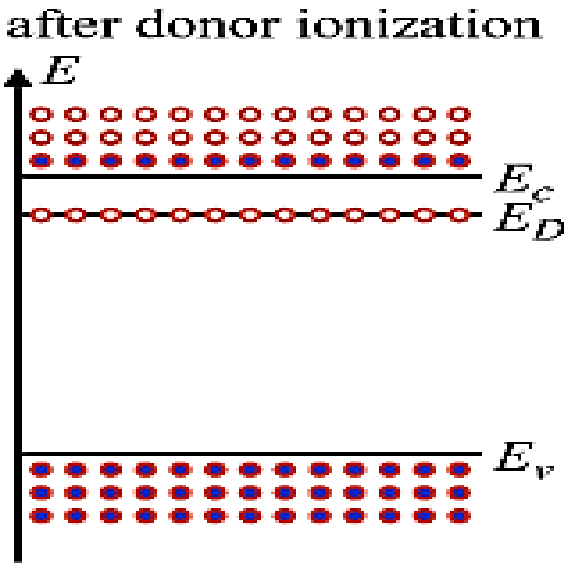

Si
Si
Si
Si
P
Si
Si
Si
Si
Free electron
Silicon atom
Sharing of electrons
Phosphorus atom


Valence electrons
Covalent bonds
+4
+4
+4
+4
+5
+4
+4
+4
+4
Free electron donated by impurity atom
Pentavalent impurity atom (donor)
Silicon atoms

Valence band: is the highest energy band which completely filled (in our example: 3s)
Conduction band: the lowest energy band which is not completely filled (in our example 3p)

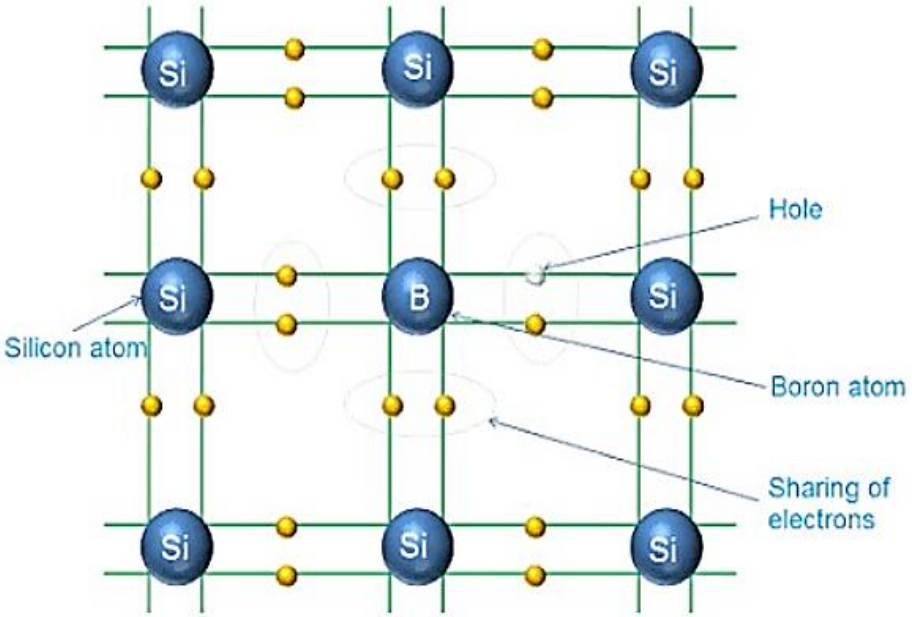


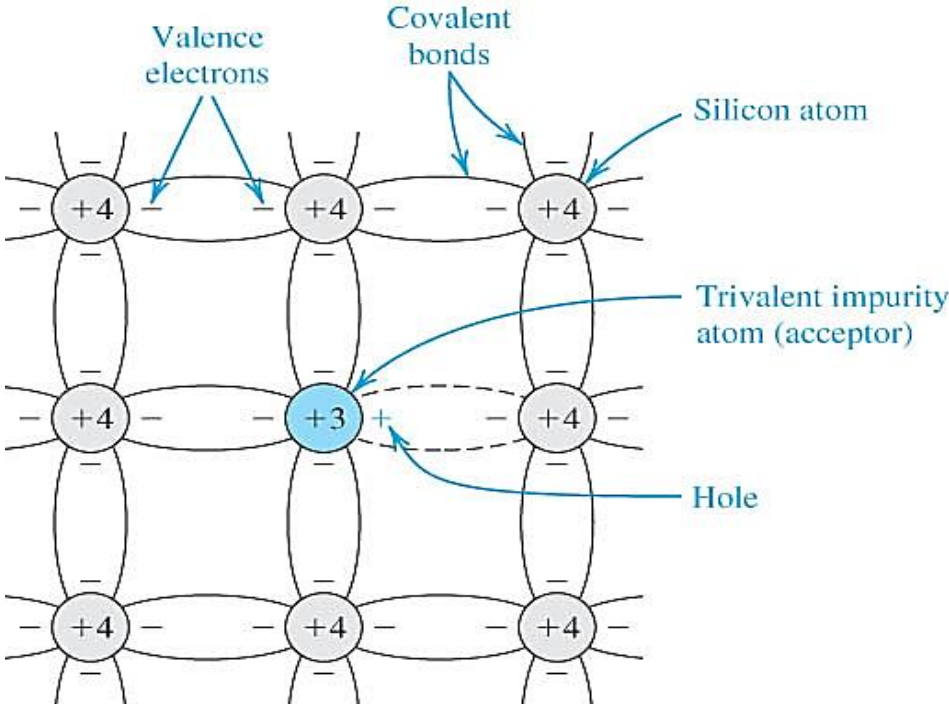


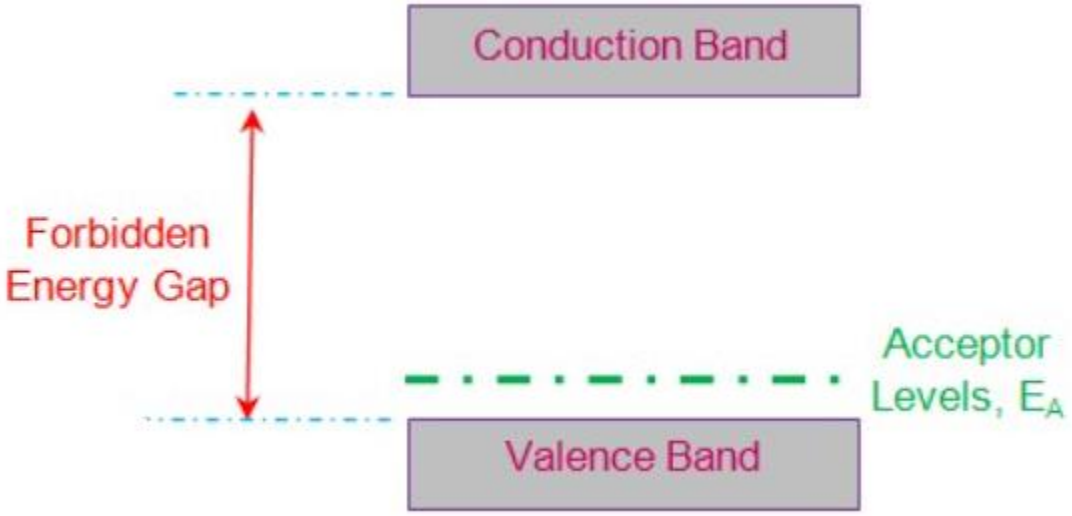


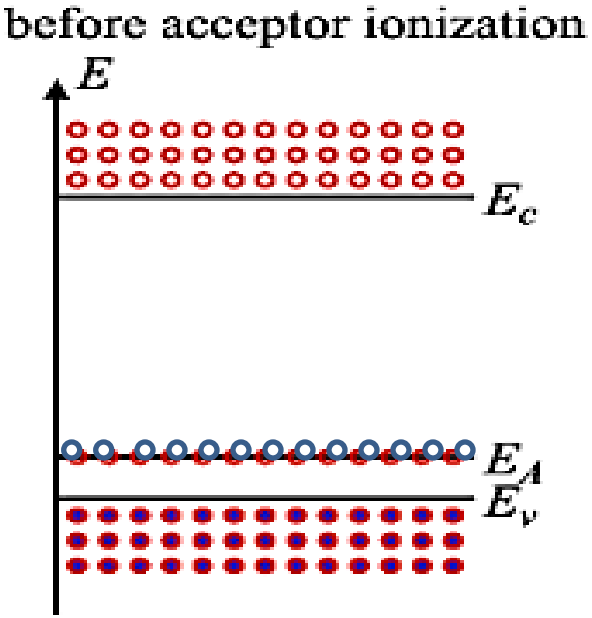
2. P- type semiconductors (Acceptors)

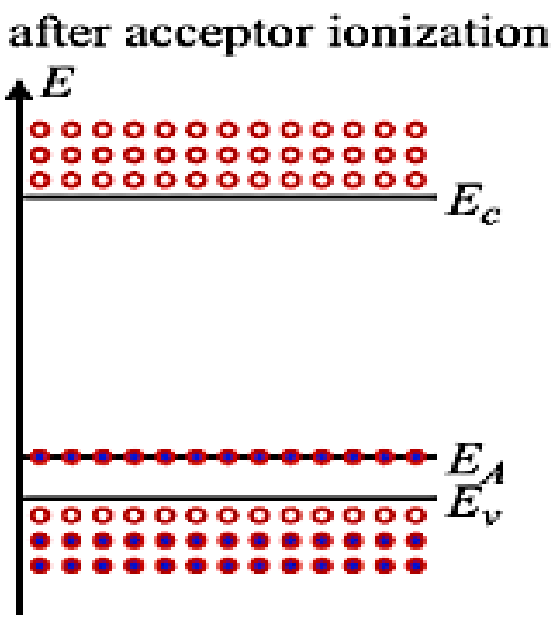
- To produce p-type semiconductor, silicon has to be doped with a **trivalent** impurity such as boron (B) and Aluminium (Al).
- trivalent impurity shares three electrons with the neighboring atom. Since an additional electron is required a free hole is appeared in the trivalent impurity (called acceptor)






Conduction Band
Forbidden
Energy Gap
Acceptor
Levels, E_A
Valence Band

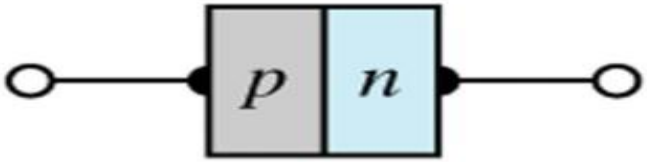

before acceptor ionization
E
E_c
E_A
E_v


after acceptor ionization
E
E_c
E_A
E_v

p-n Junctions

One end of a silicon or germanium crystal can be doped as a *p*-type material and the other end as an *n*-type material.

The result is a *p-n* junction.

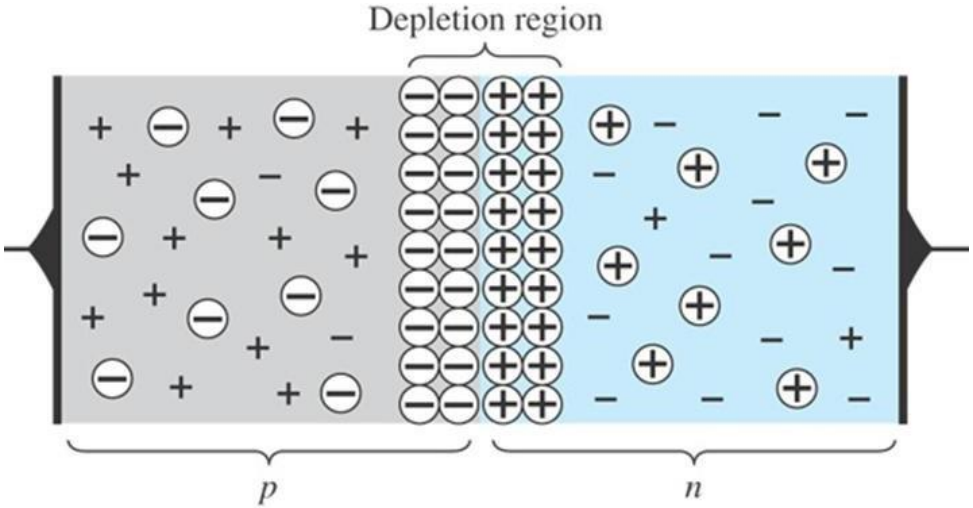


p-n Junctions

At the *p*-*n* junction, the excess conduction-band electrons on the *n*-type side are attracted to the valence-band holes on the *p*-type side.

The electrons in the *n*-type material migrate across the junction to the *p*-type material (electron flow).

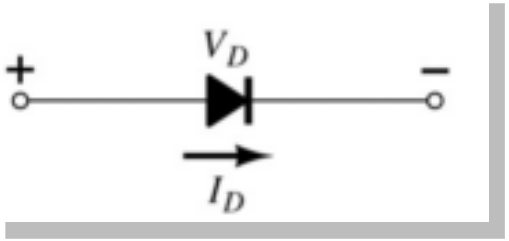
The electron migration results in a negative charge on the p-type side of the junction and a positive charge on the n-type side of the junction.



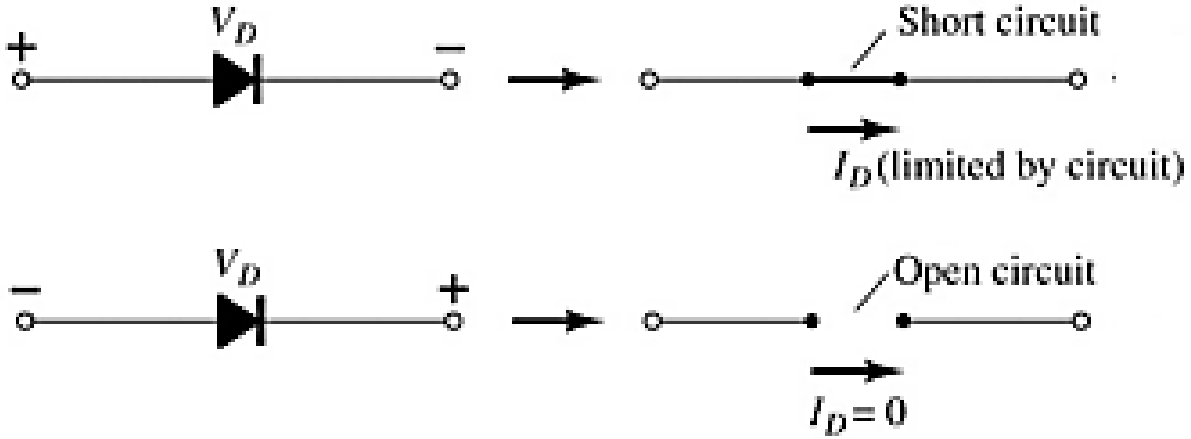
The result is the formation of a depletion region around the junction.

Diodes

The diode is a 2-terminal device.

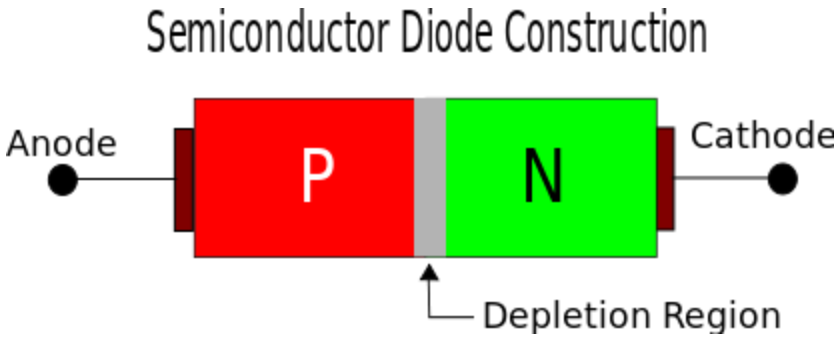


A diode ideally conducts in only one direction.

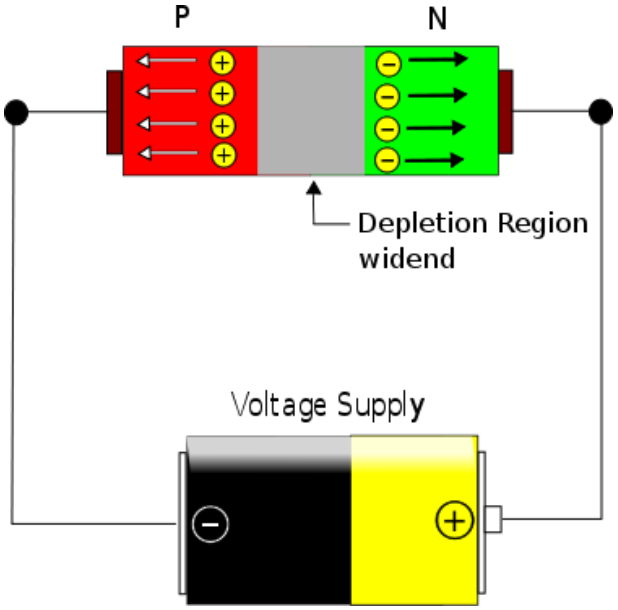


Diode Operating Conditions

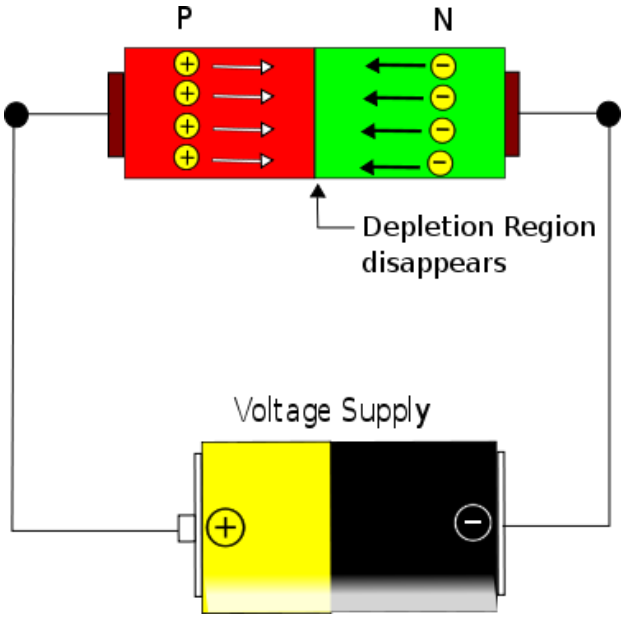
- **No bias**
- **Forward bias**
- **Reverse bias**



Reverse bias

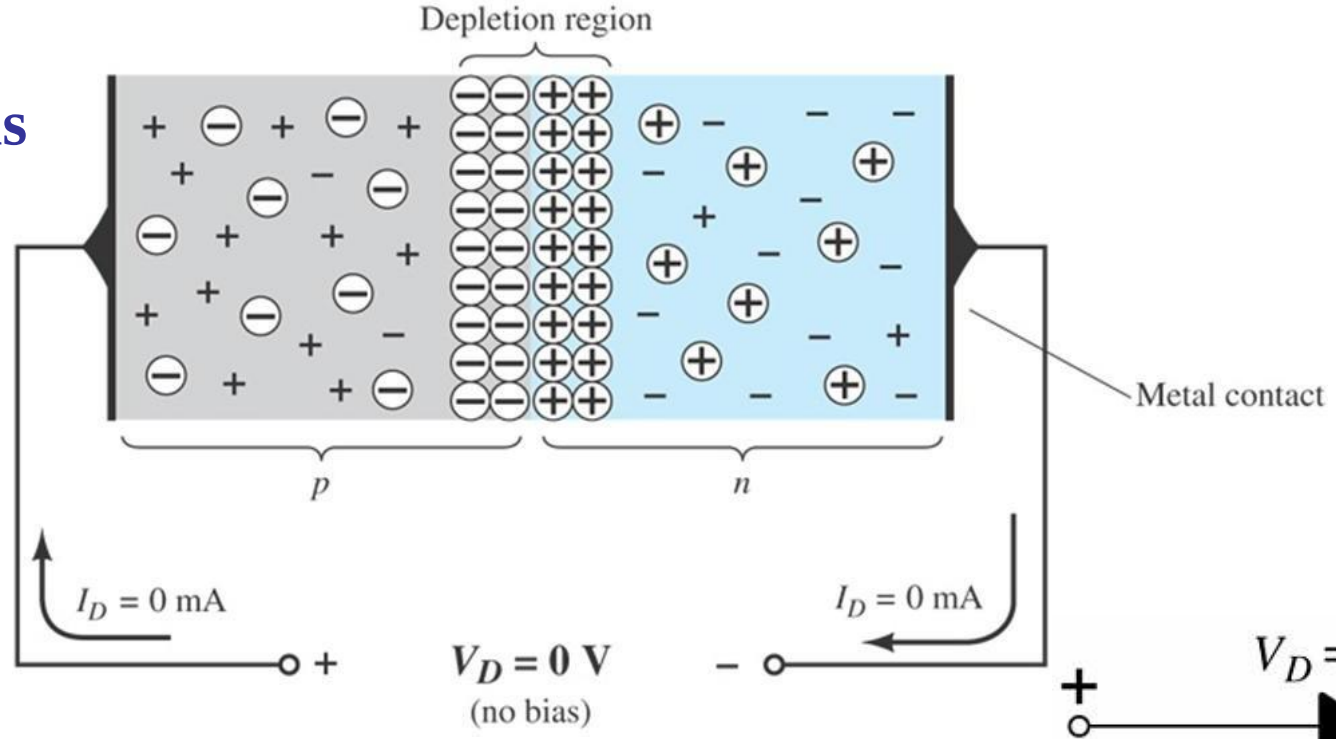


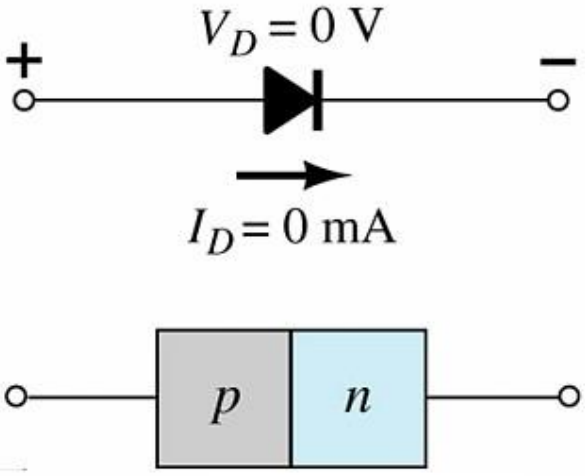
Forward bias



Diode Operating Conditions

No Bias



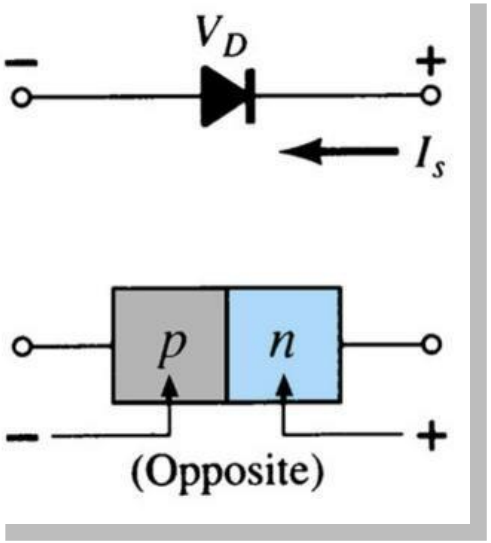


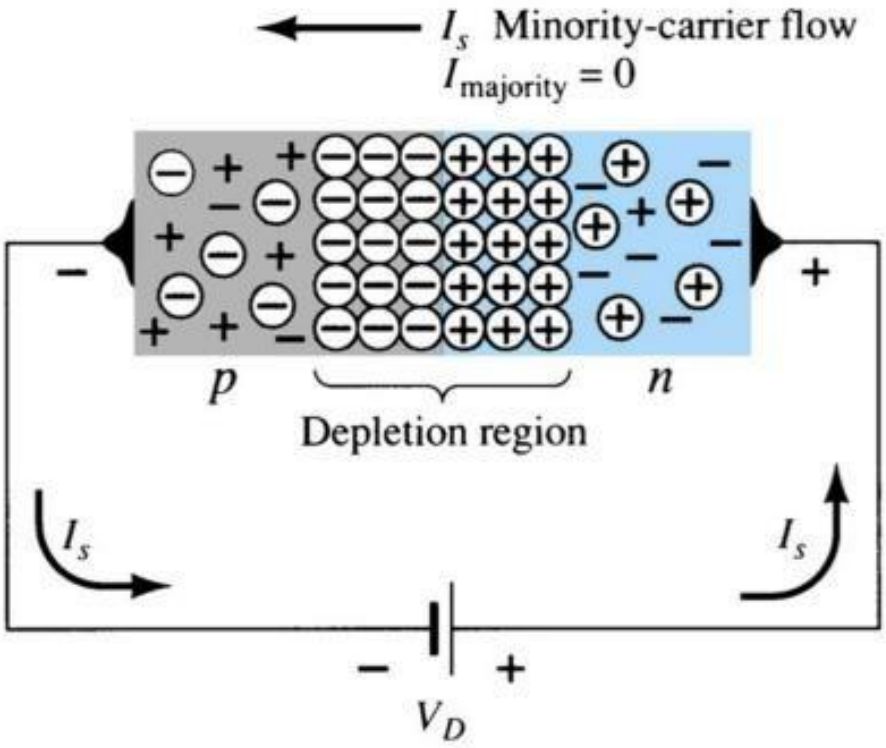
- **No external voltage is applied: $V_D = 0\,V$**
- **No current is flowing: $I_D = 0\,A$**
- **Only a modest depletion region exists**

Diode Operating Conditions

Reverse Bias

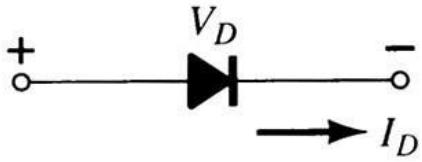
External voltage is applied across the *p*-*n* junction in the opposite polarity of the *p*- and *n*-type materials.





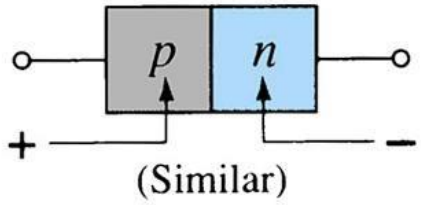
- **The reverse voltage causes the depletion region to widen.**
- **The electrons in the *n*-type material are attracted toward the positive terminal of the voltage source.**
- **The holes in the *p*-type material are attracted toward the negative terminal of the voltage source.**

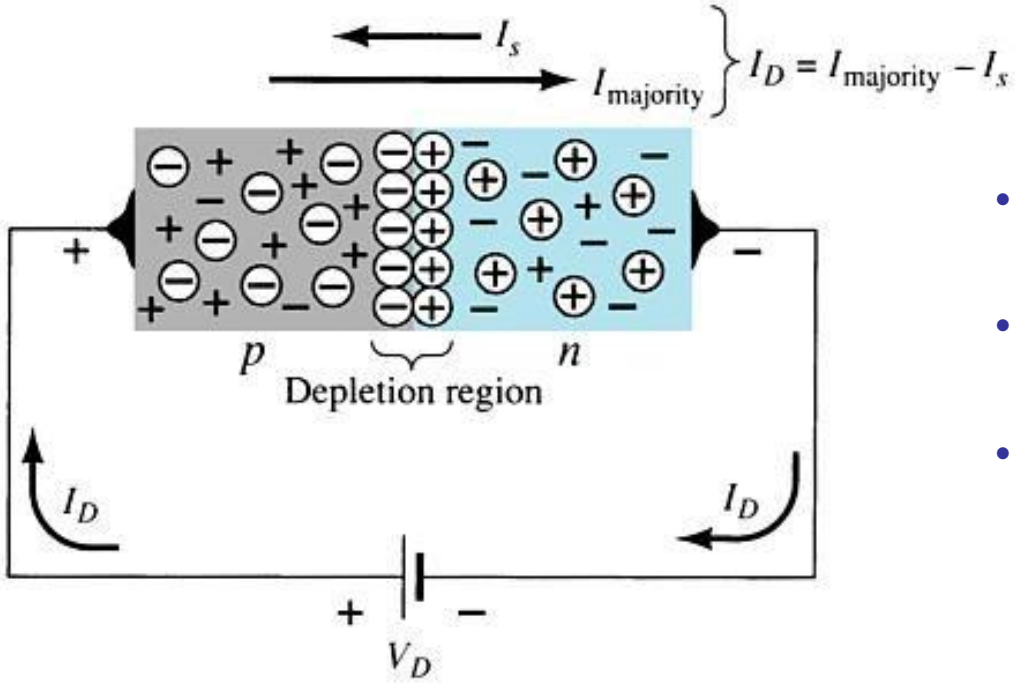
Diode Operating Conditions



Forward Bias

External voltage is applied across the *p*-*n* junction in the same polarity as the *p*- and *n*-type materials.





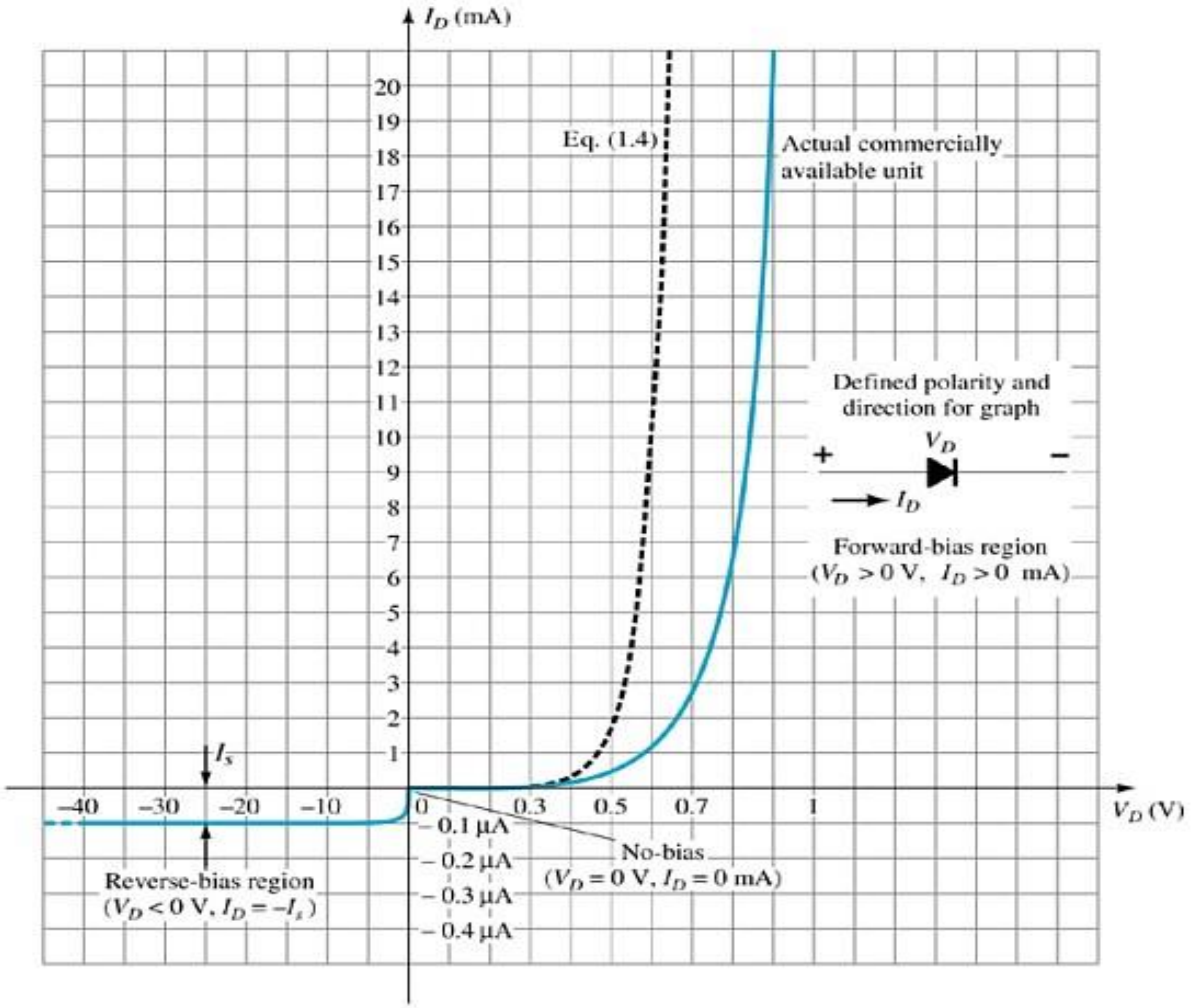
- **The forward voltage causes the depletion region to narrow.**
- **The electrons and holes are pushed toward the *p*-*n* junction.**
- **The electrons and holes have sufficient energy to cross the *p*-*n* junction.**

Actual Diode Characteristics

Note the regions for no bias, reverse bias, and forward bias conditions.

Carefully note the scale for each of these conditions.

The reverse saturation current is seldom more than a few microamperes.



Diode equation

$$I_D = I_S\left(e^{V_D/nV_T} - 1\right)$$

$$V_T = \frac{kT}{q}$$

where

$\boldsymbol{V_T}$: is called the thermal voltage.

$\boldsymbol{I_s}$: is the reverse saturation current.

$\mathbf{V_D}$: is the applied forward-bias voltage across the diode.

n : is a factor function of operation conditions and physical construction. It has range between 1 and 2. assume n=1 unless otherwise noted.

K : is Boltzman's constant =1.38 x 10^{-23}

T: is temperature in kelvins = 273+temperature in C.

q : is the magnitude of electron charge = 1.6 x 10^{-19} C.

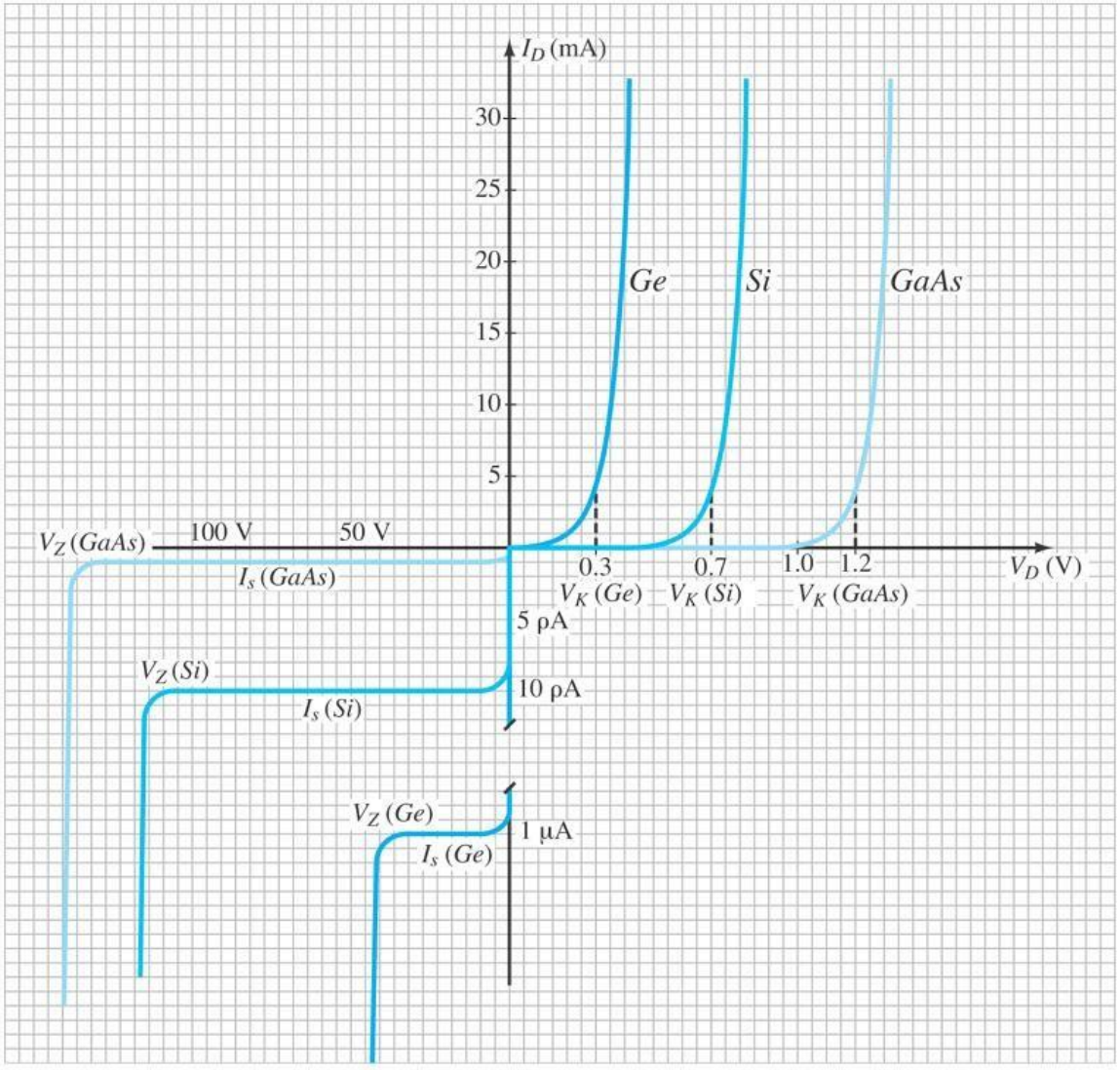
Forward Bias Voltage

The point at which the diode changes from no-bias condition to forward-bias condition occurs when the electrons and holes are given sufficient energy to cross the *p*-*n* junction. This energy comes from the external voltage applied across the diode.

The forward bias voltage required for a:

- **gallium arsenide diode ≅ 1.2 V**
- **silicon diode ≅ 0.7 V**
- **germanium diode ≅ 0.3 V**

Comparison Ge, Si, GaAs



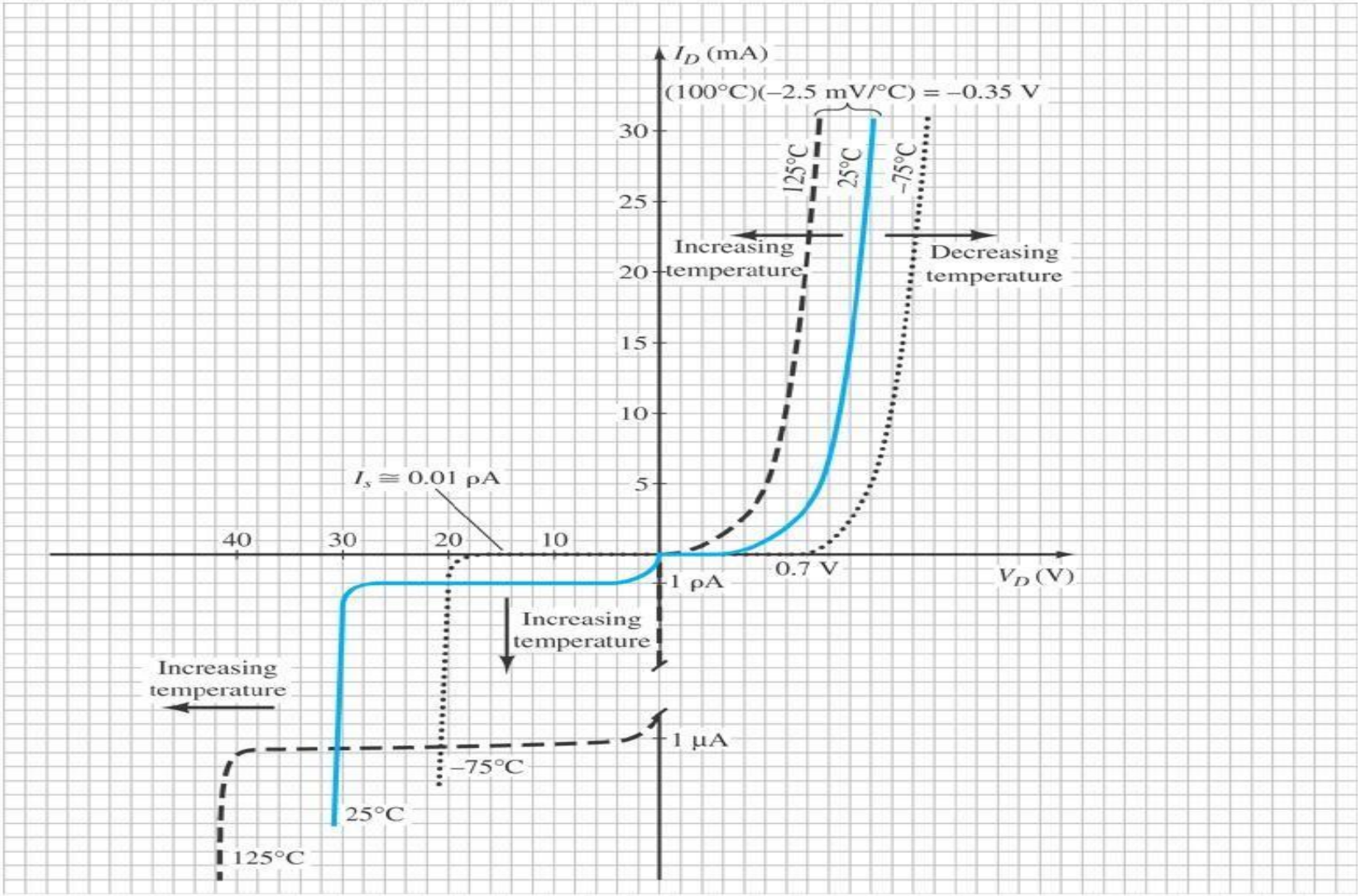
Temperature Effects

As temperature increases it adds energy to the diode.

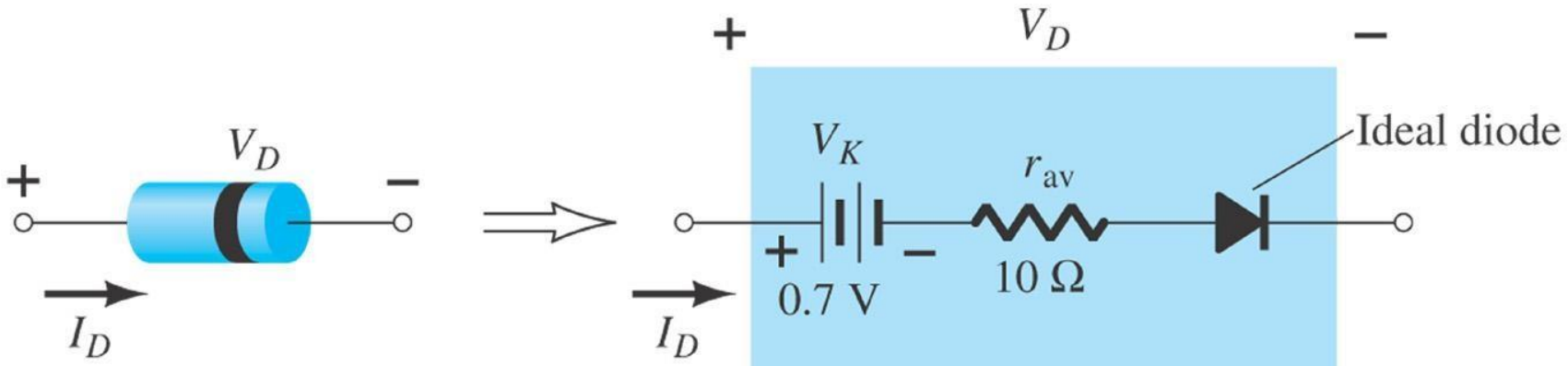
- It reduces the required forward bias voltage for forward-bias conduction.
- It increases the amount of reverse current in the reverse-bias condition.

Germanium diodes are more sensitive to temperature variations than silicon or gallium arsenide diodes.

Temperature Effects

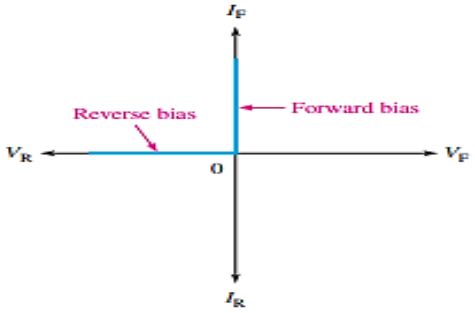
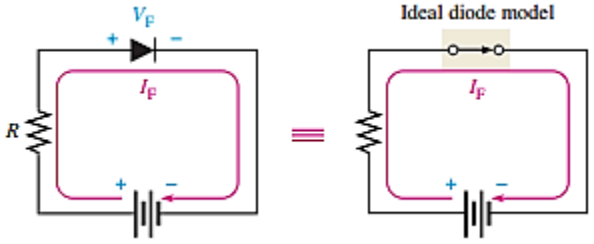
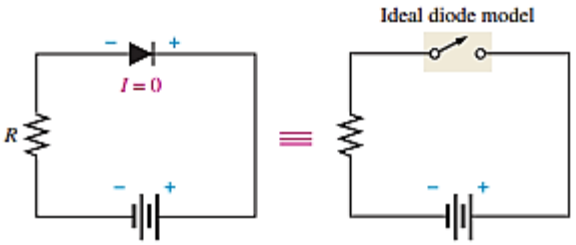


Diode Equivalent Circuit

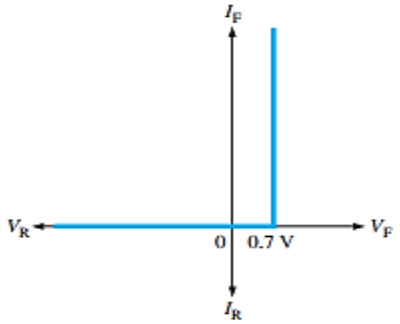
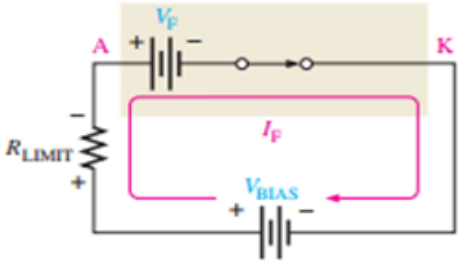
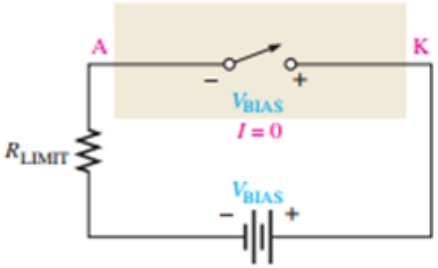


Diode Models

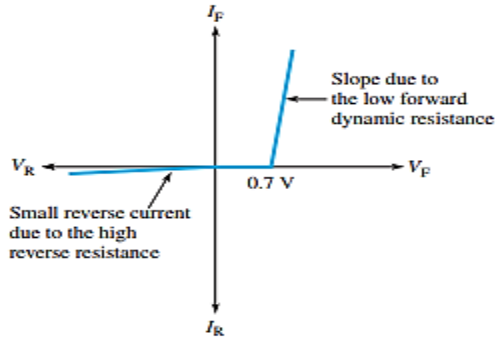
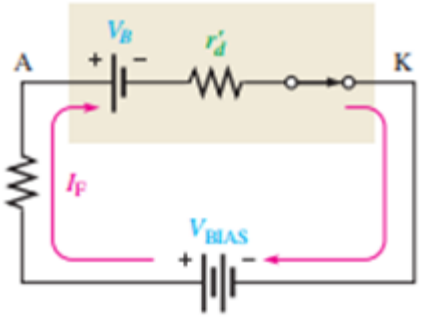
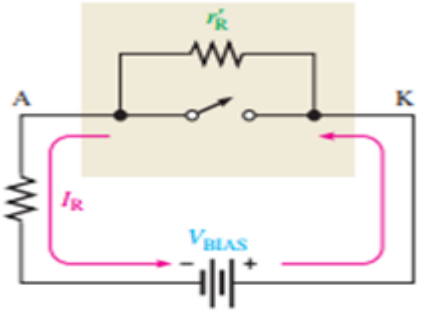
1. Ideal Model:

Characteristic curve		
Forward bias (diode represented by short circuit)		$V_F = 0\ V$ $I_F = \frac{V_{Bias}}{R_{Limiting}}$
Reverse Bias (diode represented by an open circuit)		$I_R = 0\ A$ $V_R = V_{Bias}$

2. Practical Model:

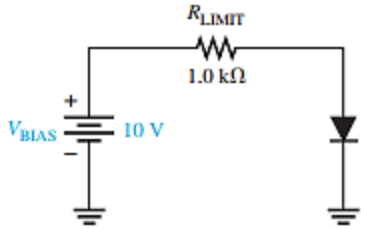
Characteristic curve		
Forward bias (diode represented by a battery of (VF=0.7 V))		$V_F = 0.7\ V$ $I_F = \frac{V_{Bias} - 0.7}{R_{Limiting}}$
Reverse Bias (diode represented by an open circuit)		$I_R = 0\ A$ $V_R = V_{Bias}$

3. Complete Model:

Characteristic curve		
Forward bias (diode represented by a battery of 0.7 V + (small forward dynamic series resistor r'_d =10 Ω)		$V_F = 0.7 + I_F r'_d$ $I_F = \frac{V_{Bias} - 0.7}{R_{Limiting} + r'_d}$
Reverse Bias (diode represented by very high resistance)		I_R=very small value (in micro) $V_F = V_{Bias} - I_R R_{Limiting}$

Example (1)

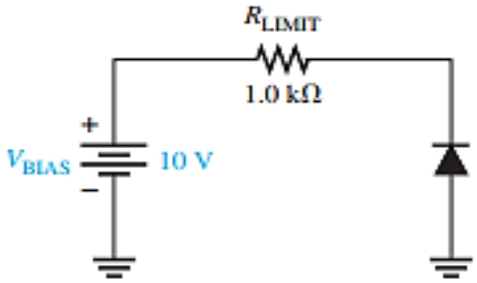
Determine the **forward voltage** and forward current for the diode in the figure below for each of the diode models. Also find the voltage across the limiting resistor in each case. Assume $r'_d = 10\Omega$ at the determined value of forward current.



Ideal Model	$V_F = 0\ V$ $I_F = \frac{10}{1 \times 10^3} = 10\ mA$ $V_R = 10\ mA \times 1K\Omega = \mathbf{10}\ V$
Practical Model	$V_F = 0.7\ V$ $I_F = \frac{10 - 0.7}{1 \times 10^3} = 9.3\ mA$ $V_R = 9.3\ mA \times 1K\Omega = \mathbf{9.3}\ V$
Complete Model	$I_F = \frac{10 - 0.7}{1 \times 10^3\Omega + 10\Omega} = 9.21\ mA$ $V_F = 0.7\ V + 9.21\ mA\ \times r'_d = 0.792\ V$ $V_R = 9.21\ mA \times 1K\Omega = \mathbf{9.21}\ V$

Example (2)

Determine the **reverse voltage** and reverse current for the diode in the figure below for each of the diode models. Also find the voltage across the limiting resistor in each case. Assume I_R = 1 µA.



Ideal Model	$I_R = 0\ V$ $V_R = V_{Bias} = 10\ V$ $V_R = 0\ V$
Practical Model	$I_R = 0\ V$ $V_R = V_{Bias} = 10\ V$ $V_R = 0\ V$
Complete Model	$I_R = 1\ \mu A$ $V_R = I_R \times R = 1\ \mu A \times 1K\Omega = 1\ mV$ $V_R = V_{BiaS} - V_R = 10\ V - 1\ mV = 9.999\ V$

Diode Specification Sheets

Data about a diode is presented uniformly for many different diodes.

1. **Forward Voltage (V_F) at a specified current and temperature**
2. **Maximum forward current (I_F) at a specified temperature**
3. **Reverse saturation current (I_R) at a specified voltage and temperature**
4. **Reverse voltage rating, PIV or PRV or V(BR), at a specified temperature**
5. **Maximum power dissipation at a specified temperature**
6. **Capacitance levels**
7. **Reverse recovery time,** t_{rr} (is the time required for a diode to stop conducting once it is switched from forward bias to reverse bias)
8. **Operating temperature range**

Diode Specification Sheets

DIFFUSED SILICON PLANAR

A — • BV . . . 125 V (MIN) @ 100 µA (BAY73)

ABSOLUTE MAXIMUM RATINGS (Note 1)

Temperatures

B

Storage Temperature Range	–65°C to +200°C
Maximum Junction Operating Temperature	+175°C
Lead Temperature	+260°C

Power Dissipation (Note 2)

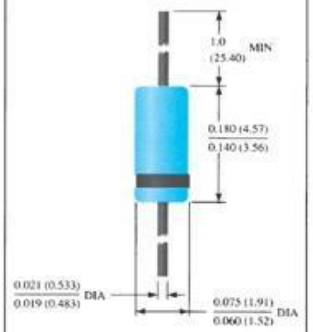
C

Maximum Total Power Dissipation at 25°C Ambient	500 mW
Linear Power Derating Factor (from 25°C)	3.33 mW/°C

Maximum Voltage and Currents

WIV	Working Inverse Voltage	BAY73	100 V
I_O	Average Rectified Current		200 mA
D — I_F	Continuous Forward Current		500 mA
i_f	Peak Repetitive Forward Current		600 mA
$i_{f(surge)}$	Peak Forward Surge Current		
	Pulse Width = 1 s		1.0 A
	Pulse Width = 1 µs		4.0 A

DO-35 OUTLINE



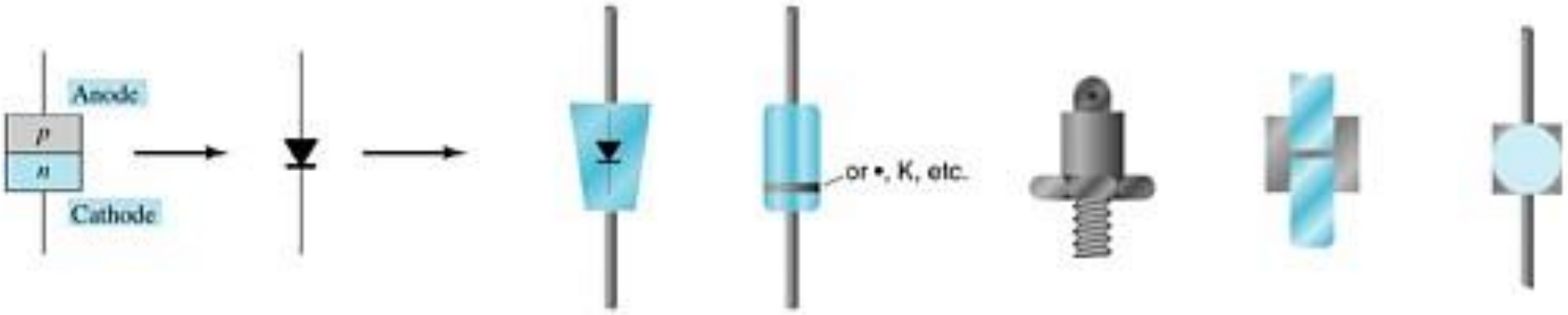
NOTES:
Copper clad steel leads, tin plated
Gold plated leads available
Hermetically sealed glass package
Package weight is 0.14 gram

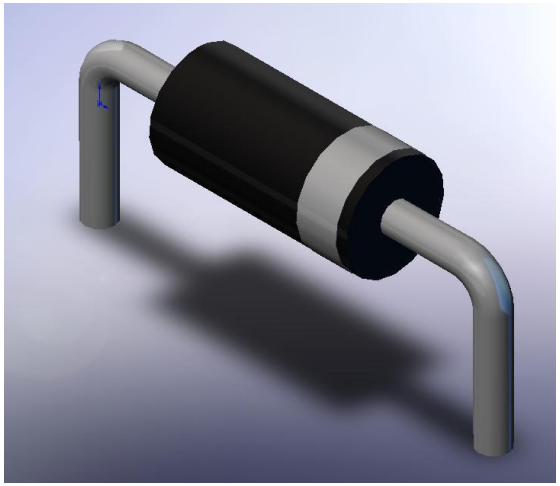
ELECTRICAL CHARACTERISTICS (25°C Ambient Temperature unless otherwise noted)

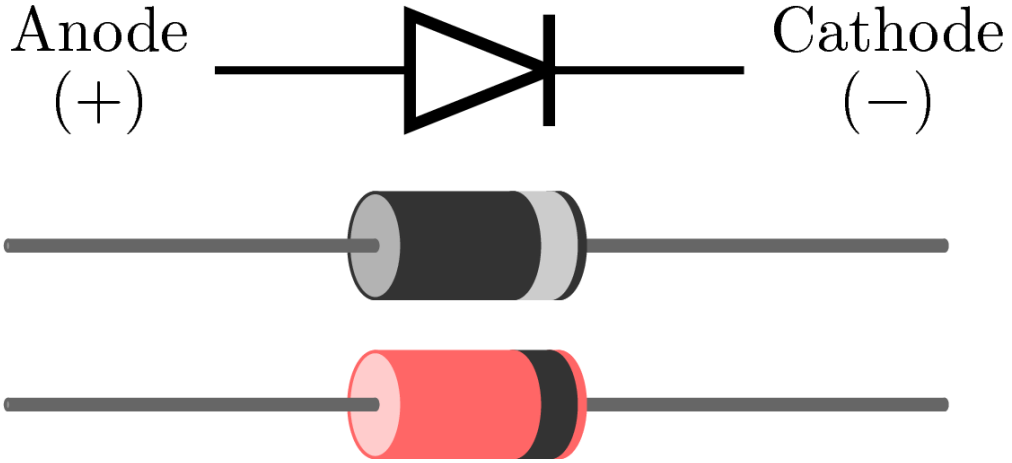
SYMBOL	CHARACTERISTIC	BAY73 MIN	BAY73 MAX	UNITS	TEST CONDITIONS
E — V_F	Forward Voltage	0.85	1.00	V	I_F = 200 mA
		0.81	0.94	V	I_F = 100 mA
		0.78	0.88	V	I_F = 50 mA
		0.69	0.80	V	I_F = 10 mA
		0.67	0.75	V	I_F = 5.0 mA
		0.60	0.68	V	I_F = 1.0 mA
F — I_R	Reverse Current		500	nA	V_R = 20 V, T_A = 125°C
			1.0	µA	V_R = 100 V, T_A = 125°C
			0.2	nA	V_R = 20 V, T_A = 25°C
			0.5	nA	V_R = 100 V, T_A = 25°C
BV	Breakdown Voltage	125		V	I_R = 100 µA
G — C	Capacitance		8.0	pF	V_R = 0, f = 1.0 MHz
H — t_{rr}	Reverse Recovery Time		3.0	µs	I_F = 10 mA, V_R = 35 V, R_L = 1.0 to 100 kΩ, C_L = 10 pF, JAN 256

NOTES
1 These ratings are limiting values above which the serviceability of the diode may be impaired.
2 These are steady state limits. The factory should be consulted on applications involving pulses or low duty-cycle operation.

Diode Symbol and Packaging

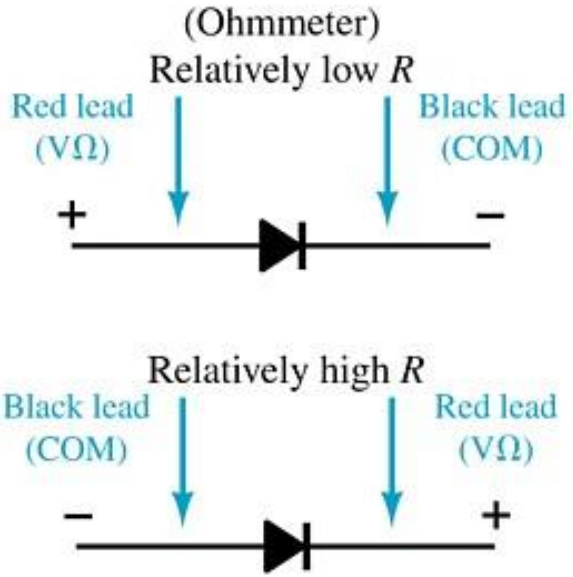


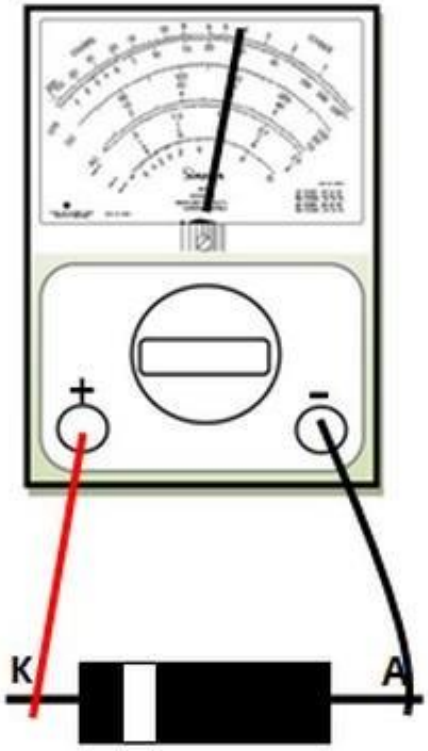


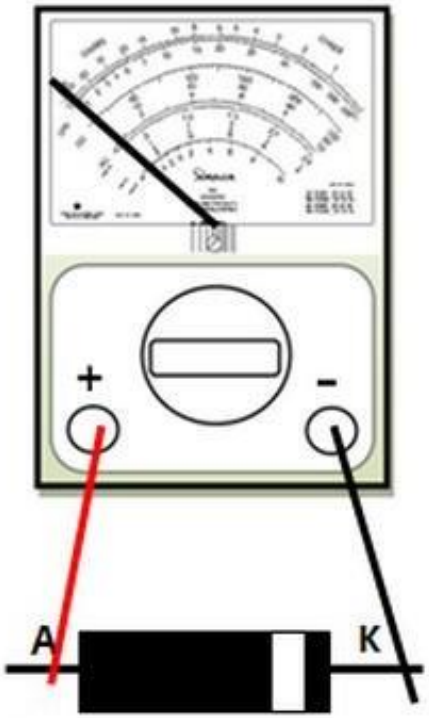


Diode Testing - Ohmmeter

An ohmmeter set on a low Ohms scale can be used to test a diode. The diode should be tested out of circuit.







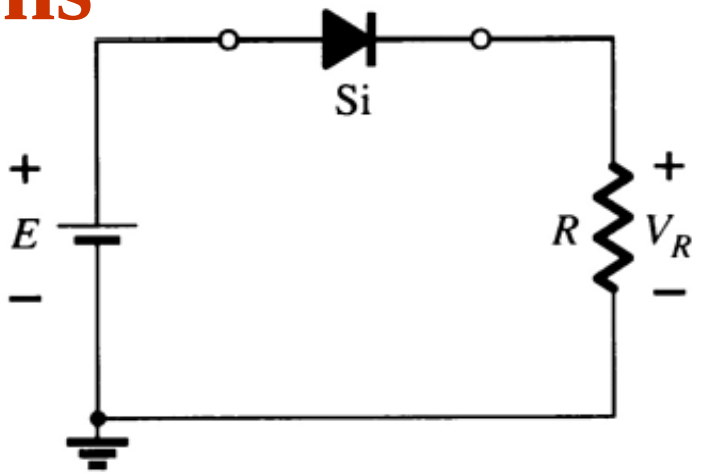
Diode Analysis

Diode Configurations

- The forward resistance of the diode is usually so small compared to the other series elements of the network that it can be ignored.
- In general, a diode is in the "on" state if the current established by the applied sources is such that its direction matches that of the arrow in the diode symbol, and $V_D \geq 0.7V$ for silicon, $V_D \geq 0.3V$ for germanium, and $V_D \geq 1.2V$ for gallium arsenide.
- You may assume the diode is "on", and then find the current in the diode. If the current flows into the positive terminal of the diode, then the assumption is right, otherwise, the diode is "off".

Series Diode Configurations

Forward Bias

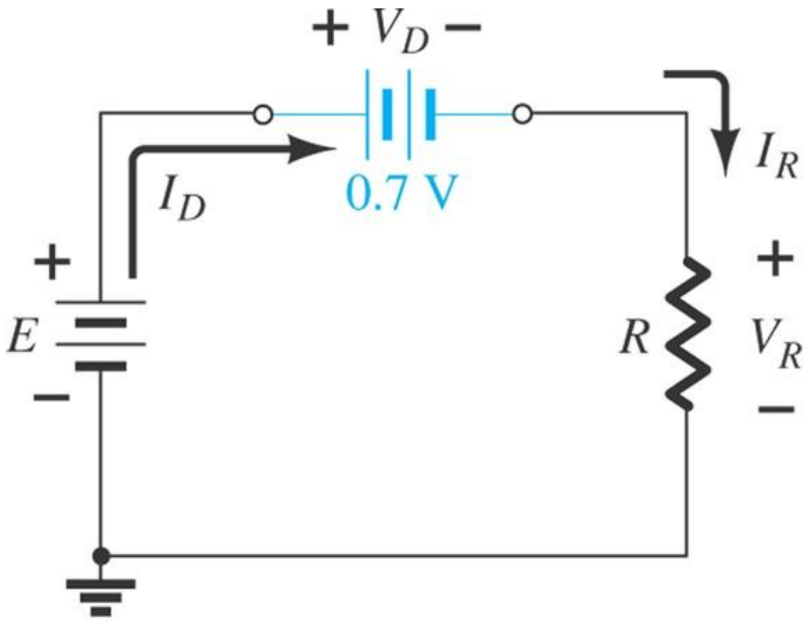


Constants

- **Silicon Diode: $V_D = 0.7\,V$**
- **Germanium Diode: $V_D = 0.3\,V$**

Analysis (for silicon)

- **$V_D = 0.7$ V (or $V_D = E$ if $E < 0.7$ V)**
- **$V_R = E - V_D$**
- **$I_D = I_R = I_T = V_R / R = (E\text{-}V_D) / R$**



Equivalent circuit for the “on” diode

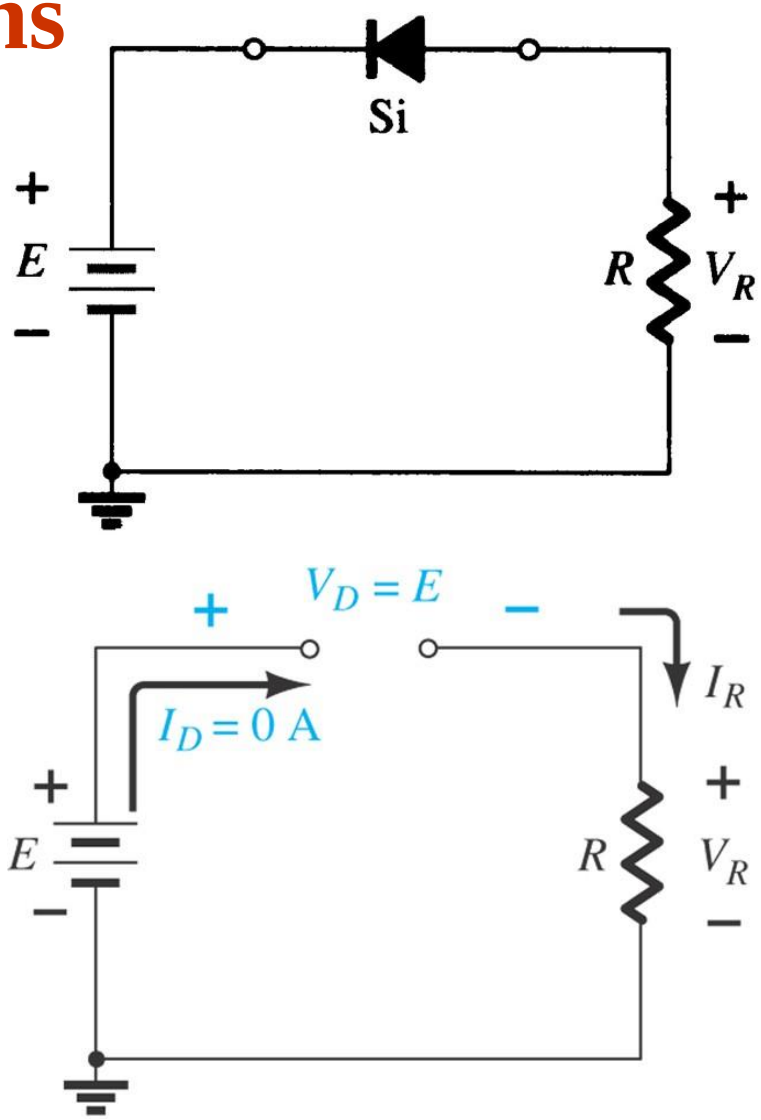
Series Diode Configurations

Reverse Bias

Diodes ideally behave as open circuits

Analysis

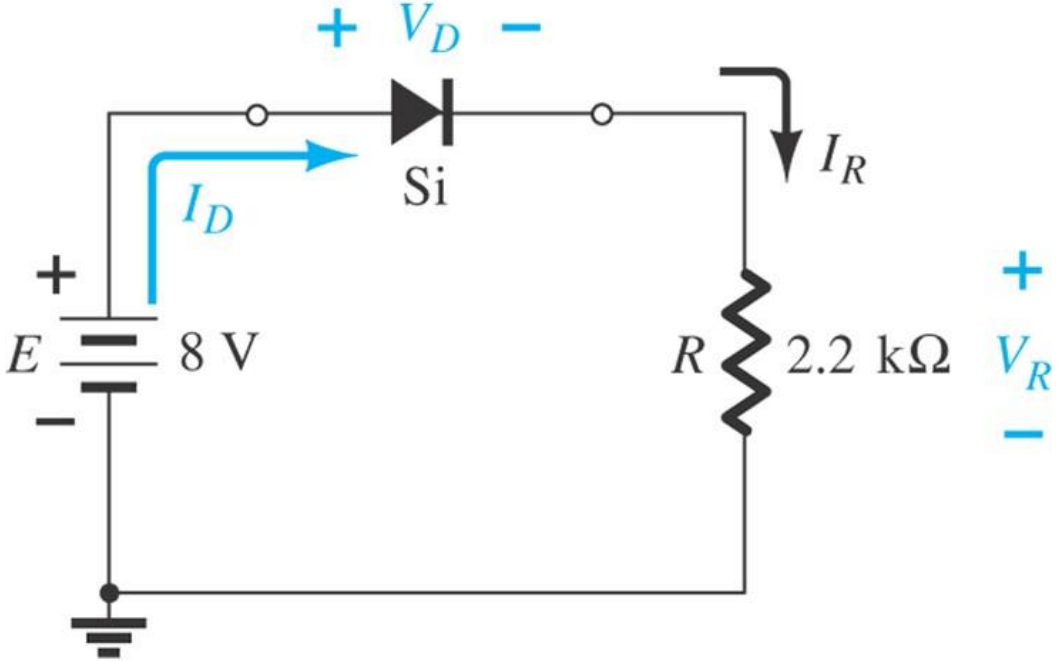
- $V_D = E$
- $V_R = 0$ V
- $I_D = 0$ A



Equivalent circuit for the “off” diode

Example

- Determine V_D, V_R and I_D.



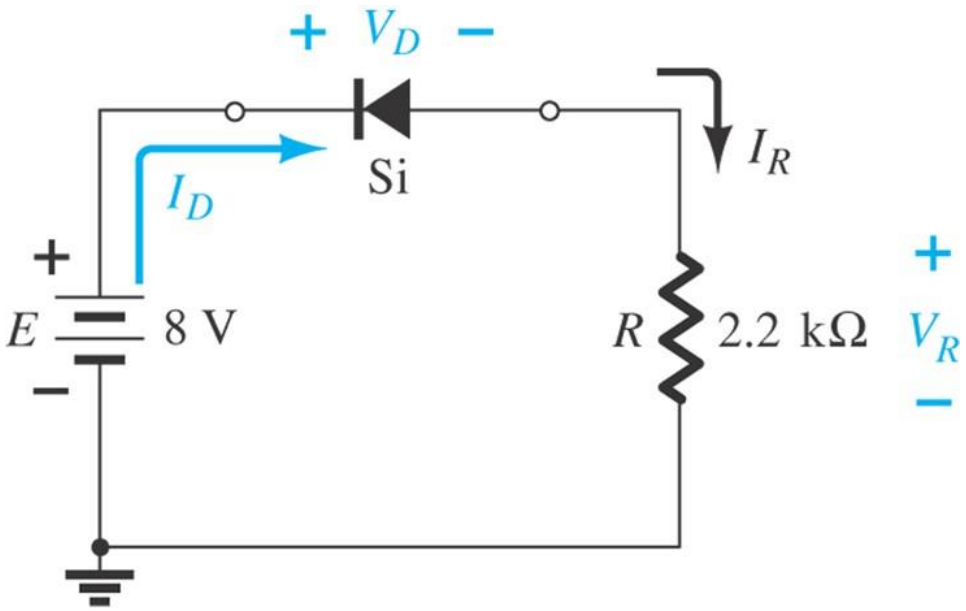
$$V_D = 0.7\,\text{V}$$

$$V_R = E - V_D = 8\,\text{V} - 0.7\,\text{V} = 7.3\,\text{V}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{7.3\,\text{V}}{2.2\,\text{k}\Omega} \cong 3.32\,\text{mA}$$

Example

Determine V_D, V_R and I_D.

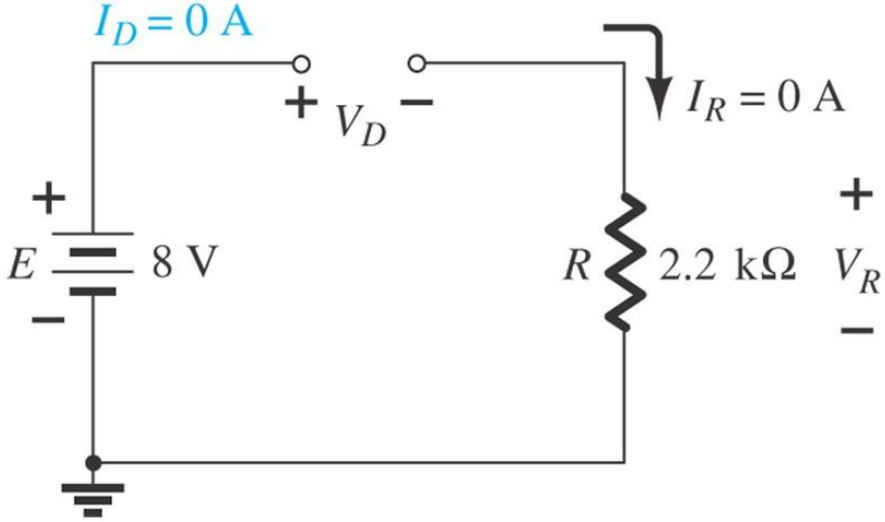


Solution

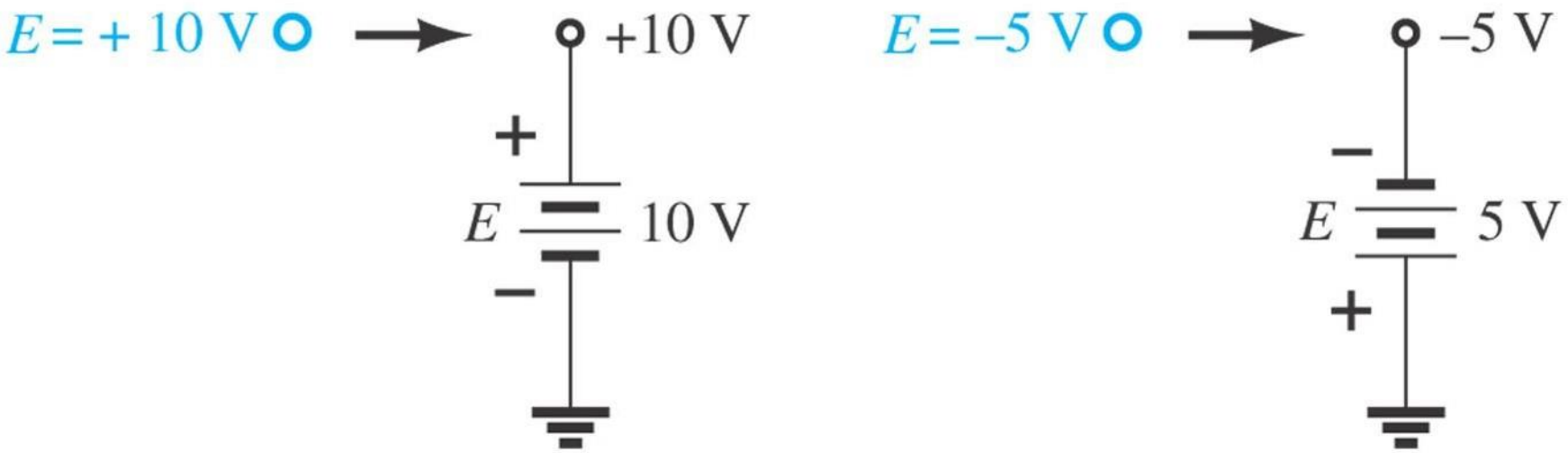
$I_D = 0$ A due to the open circuit.
$V_R = I_R R$, we have $V_R = (0)R = 0$ V.

Applying Kirchhoff's voltage law around the closed loop:

$E - V_D - V_R = 0$
and $V_D = E - V_R = E - 0 = E = 8$ V



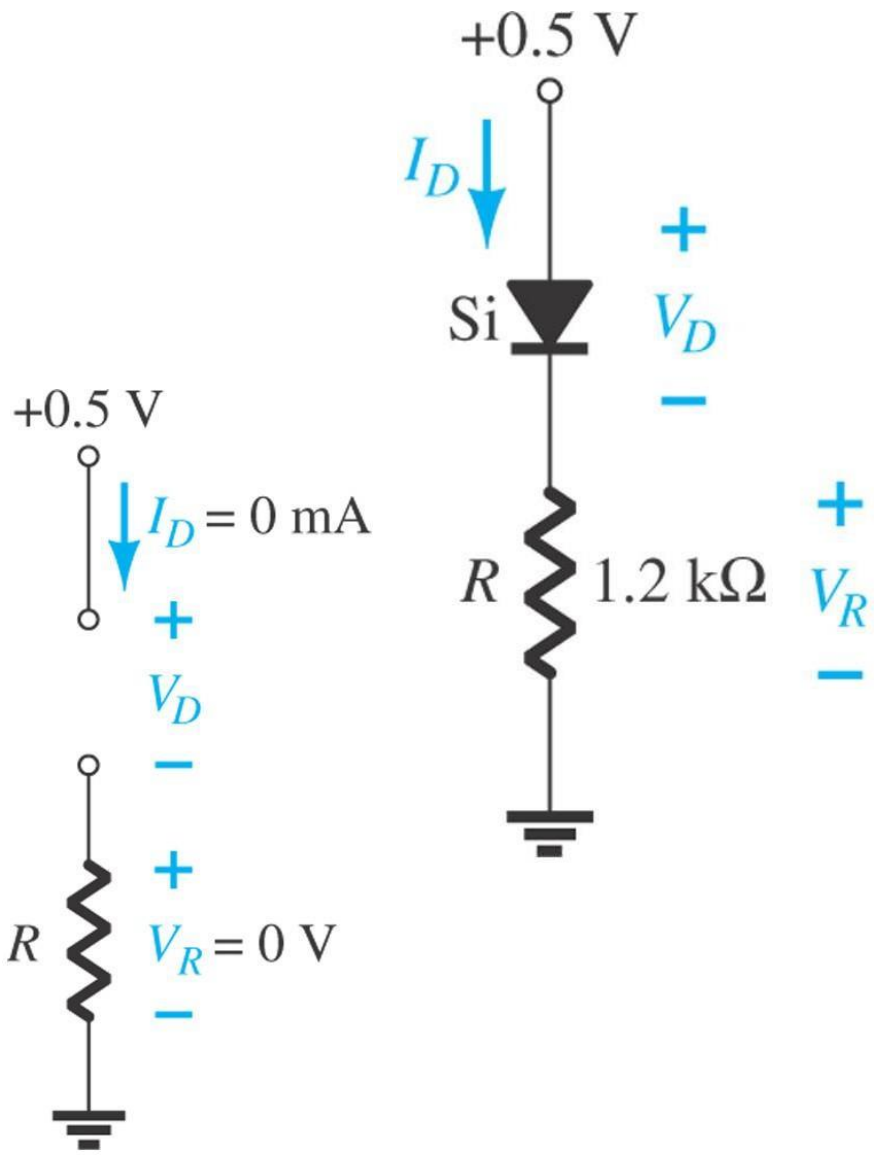
Source Notation



Example

Determine V_D, V_R and I_D.

Solution



The level of the applied voltage is insufficient (غير كافي) to turn on the silicon diode (applied voltage must be greater than 0.7 for Si but the given is 0.5)

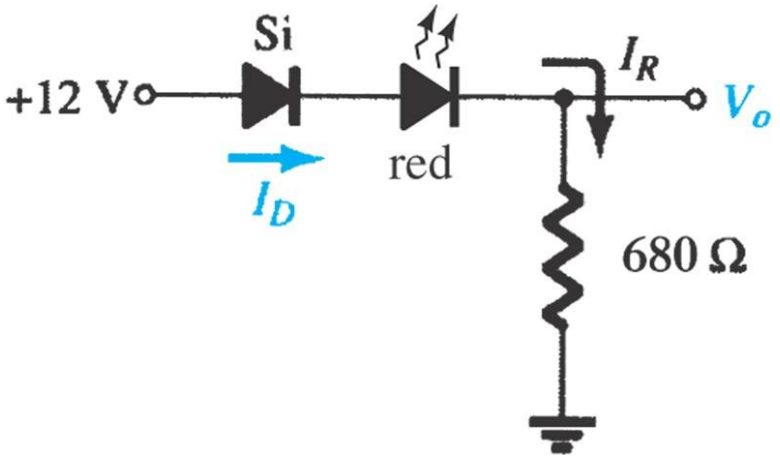
diode is open circuit

$I_D = 0$ A
$V_R = I_R R = I_D R = (0\text{ A})\ 1.2\ \text{k}\Omega = 0$ V
$V_D = E = 0.5$ V

Example

Determine V_o and I_D. The forward bias voltage for red LED is 1.8 V.

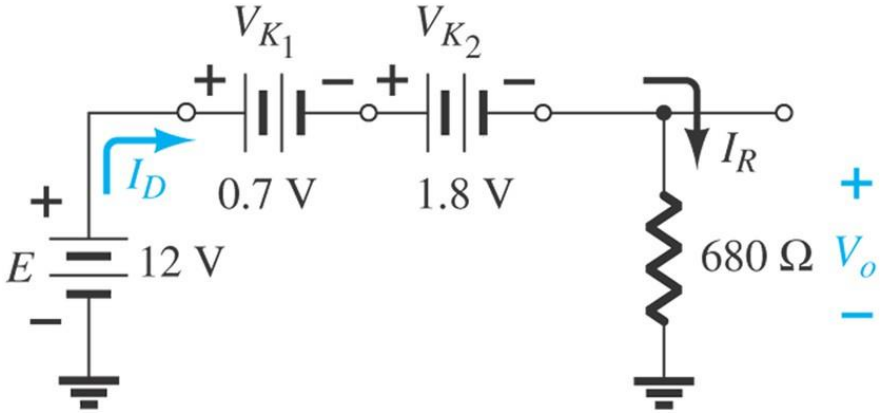
Solution



E = 12 V > (0.7 V + 1.8 V= 2.5 V)

$$V_o = E - V_{K_1} - V_{K_2} = 12\,\text{V} - 2.5\,\text{V} = \mathbf{9.5\,V}$$

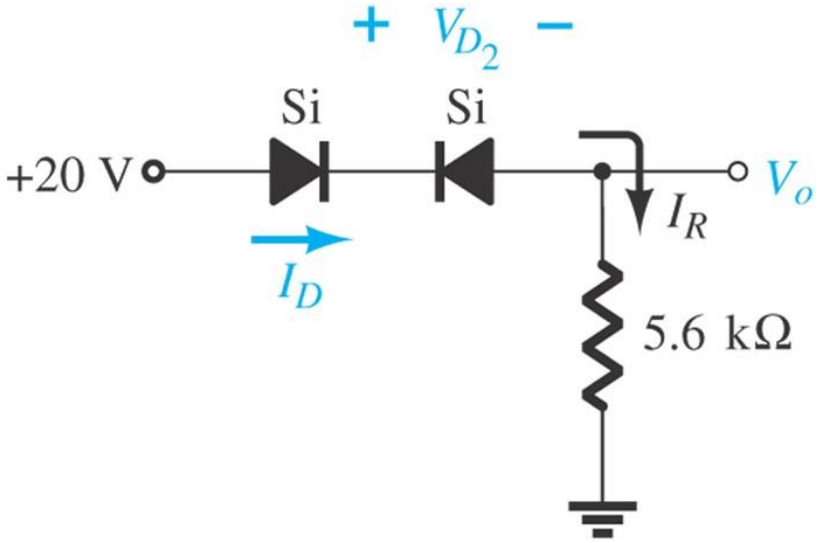
$$I_D = I_R = \frac{V_R}{R} = \frac{V_o}{R} = \frac{9.5\,\text{V}}{680\,\Omega} = \mathbf{13.97\,mA}$$



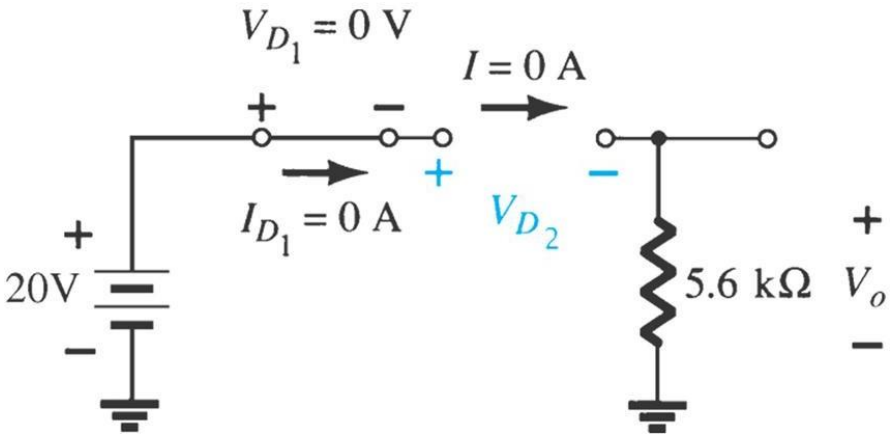
Example

Determine I_D, V_{D2} and V_o.

Solution

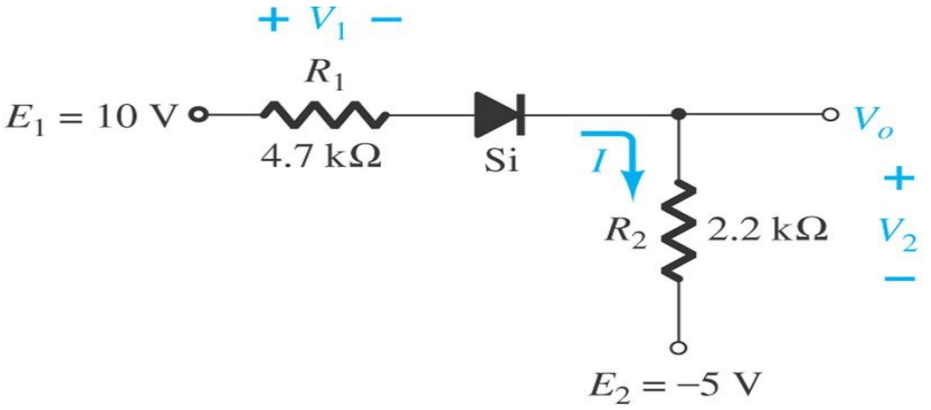


D_1 is On
D_2 is Off
$I_D = 0$ A
$Vo = I_R R = I_D R = (0\ A)R = 0$ V
$V_{D2} = V_{open\ circuit} = E = 20$ V
Applying Kirchhoff's voltage:
$E - V_{D1} - V_{D2} - V_o = 0$
$V_{D2} = E - V_{D1} - Vo = 20\ V - 0 - 0 = 20$ V
And $V_o = 0$ V



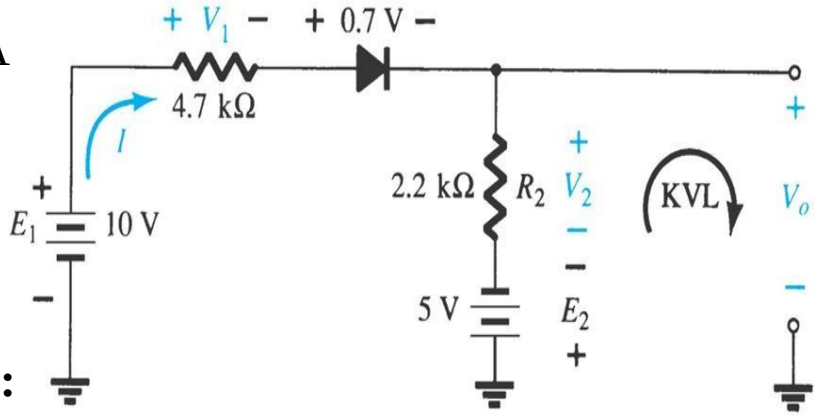
Example

Determine I, V_1, V_2 and V_o



Solution

$$I = \frac{E_1 + E_2 - V_D}{R_1 + R_2} = \frac{10\,\text{V} + 5\,\text{V} - 0.7\,\text{V}}{4.7\,\text{k}\Omega + 2.2\,\text{k}\Omega} = \frac{14.3\,\text{V}}{6.9\,\text{k}\Omega} = 2.07\ \text{mA}$$



$$V_1 = IR_1 = (2.07\ \text{mA})(4.7\ \text{k}\Omega) = \mathbf{9.73\ V}$$

$$V_2 = IR_2 = (2.07\ \text{mA})(2.2\ \text{k}\Omega) = \mathbf{4.55\ V}$$

Applying Kirchhoff's voltage law to the second loop:

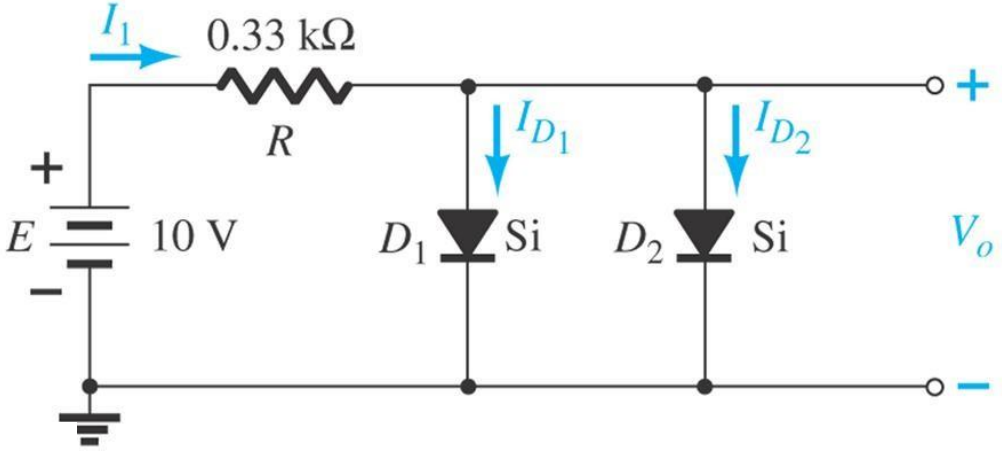
$$-E_2 + V_2 - V_o = 0$$

$$V_o = V_2 - E_2 = 4.55\ \text{V} - 5\ \text{V} = \mathbf{-0.45\ V}$$

Negative sign of V_o means that it has an **opposite polarity**

Parallel and Series-Parallel Configurations

Determine V_O, I_1, I_{D1}, and I_{D2}

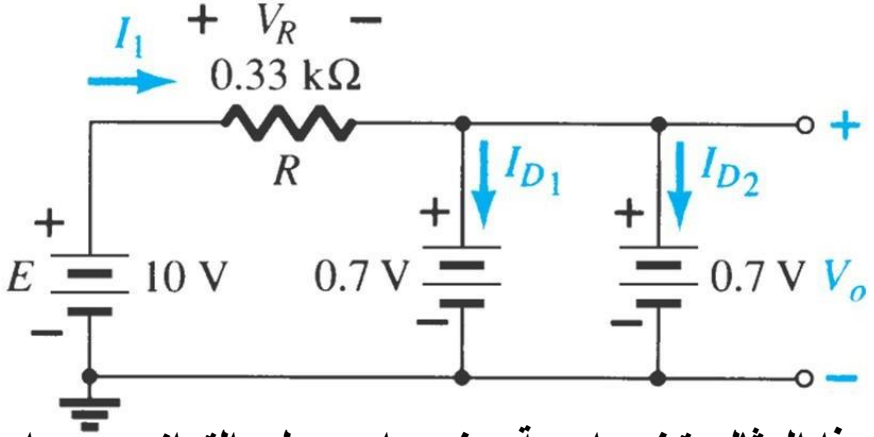


Solution

V_o= 0.7 V

$$I_1 = \frac{V_R}{R} = \frac{E - V_D}{R} = \frac{10\,\text{V} - 0.7\,\text{V}}{0.33\,\text{k}\Omega} = \mathbf{28.18\,mA}$$

Assume that the two diodes have the same characteristics:

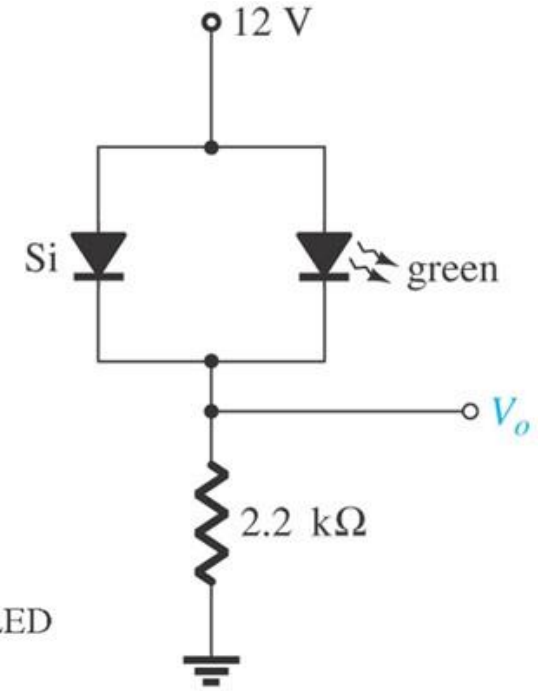


$$I_{D_1} = I_{D_2} = \frac{I_1}{2} = \frac{28.18\,\text{mA}}{2} = \mathbf{14.09\,mA}$$

في هذا المثال يتضح اهمية وضع دايود علي التوازي مع دايود اخر حيث في هذا المثال كان التيار الخارج من المصدر حوالي 28.18 ميللي امبير. لو افترضنا ان التيار المسموح به في الدايود الاول كان حوالي 20 ميللي امبير من الداتا شيت هذا معناه ان الدايود سيحترق ولهذا عندما قمنا بوضع دايود اخر علي التوازي بحساب التيار وجدناه حوالي 14 ميللي امبير اي اقل من الحد المسموح به وبها فقد قمنا بعمل حمايه للنظام بالحد من التيار الكهربي

Example

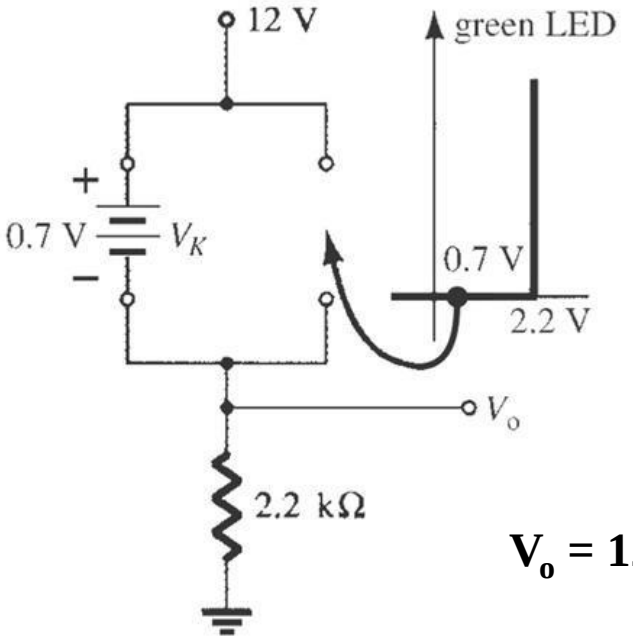
Determine the voltage V_o.



Solution

بالرغم من ان شكل الدايره يجعلنا نقول ان ال (2 diodes are ON) لكن هذا لايتماشي مع القواعد الاساسيه للدوائر الكهربيه في حالة التوازي لان شرط التوصيل التوازي يجعل الجهود متساويه وطبعا جهد (V_D) الدايود العادي يختلف عن ال(LED)

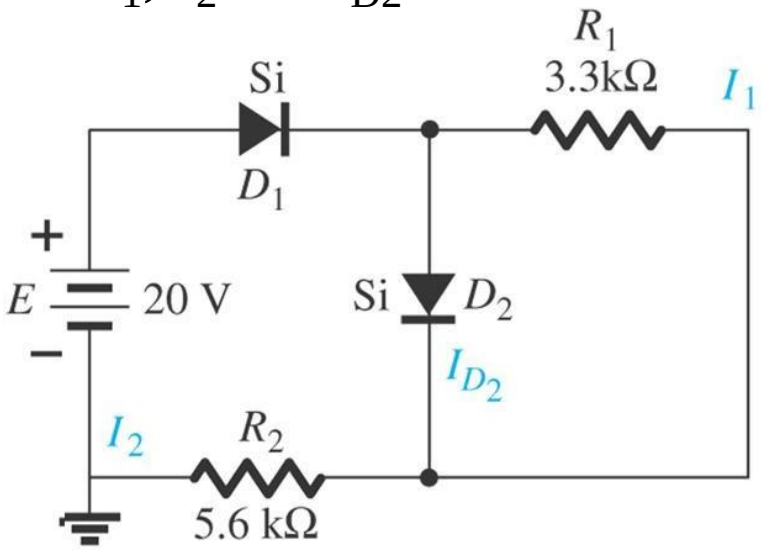
التفسير الهندسي المناسب هو ان جهد المصدر لا يصل الي قيمته العظمي من اول لحظه لكن يحتاج لحظات صغيره جدا جدا (μ sec) ولهذا فان الدايود العادي سيكون (ON) قبل ال (LED) مما يجعل الجهد علي ال (LED) يساوي ال (0.7) نظرا للتوازي وهو جهد اقل من الجهد المطلوب لجعل ال (Green LED) مفتوح لذلك فان ال (LED) سيكون (open circuit)



V_o = 12 V - 0.7 V = 11.3 V

Example

Determine the currents I_1, I_2 and I_{D2}



$$I_1 = \frac{V_{K_2}}{R_1} = \frac{0.7\text{ V}}{3.3\text{ k}\Omega} = \mathbf{0.212\text{ mA}}$$

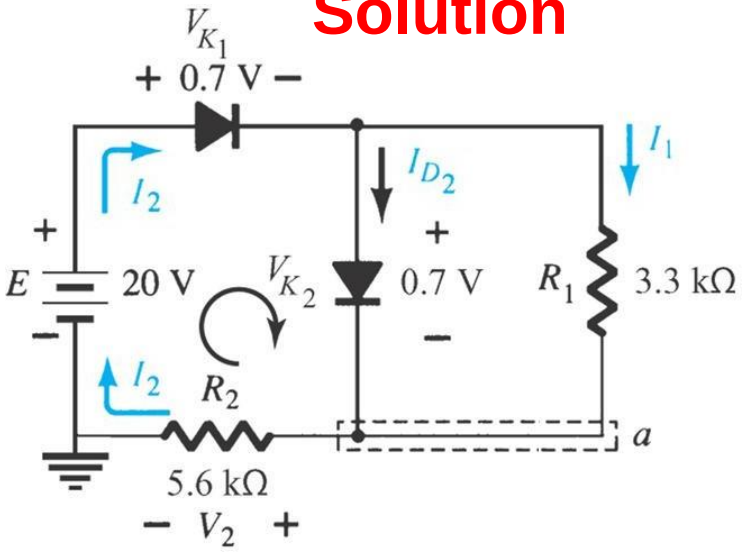
$$-V_2 + E - V_{K_1} - V_{K_2} = 0$$

$$V_2 = E - V_{K_1} - V_{K_2} = 20\text{ V} - 0.7\text{ V} - 0.7\text{ V} = \mathbf{18.6\text{ V}}$$

$$I_2 = \frac{V_2}{R_2} = \frac{18.6\text{ V}}{5.6\text{ k}\Omega} = \mathbf{3.32\text{ mA}}$$

$$I_{D_2} = I_2 - I_1 = 3.32\text{ mA} - 0.212\text{ mA} \cong 3.11\text{ mA}$$

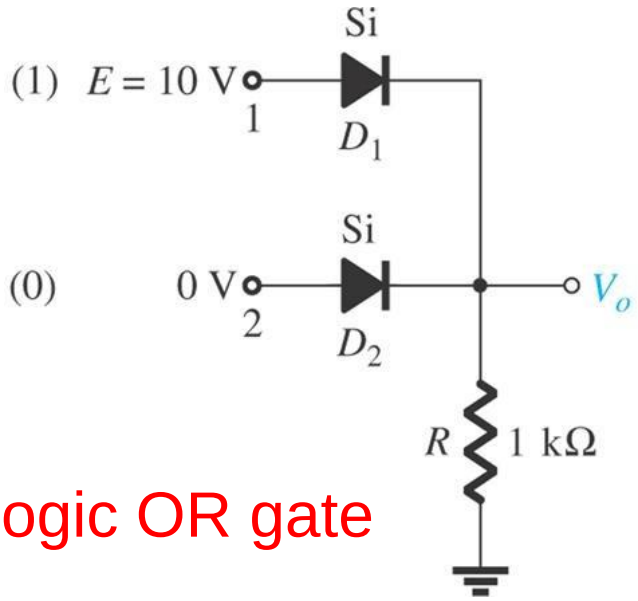
Solution



OR Gates

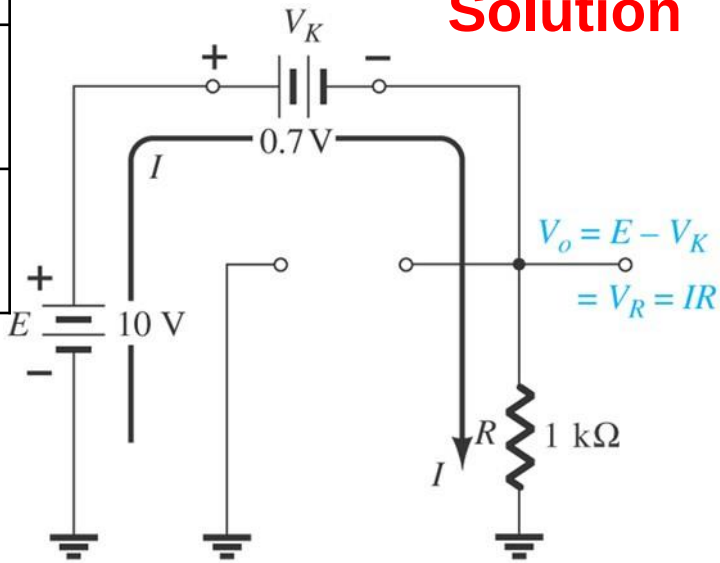
Determine V_o

Case	D1 (Si)	D2 (Si)	V_o
V_1=10 V_2=10	Forward	Forward	V=9.3 (level 1)
V_1=10 V_2=0	Forward	Reverse	V=9.3 (level 1)
V_1=0 V_2=10	Reverse	Forward	V=9.3 (level 1)
V_1=0 V_2=0	Reverse	Reverse	V=0 (level 0)



Logic OR gate

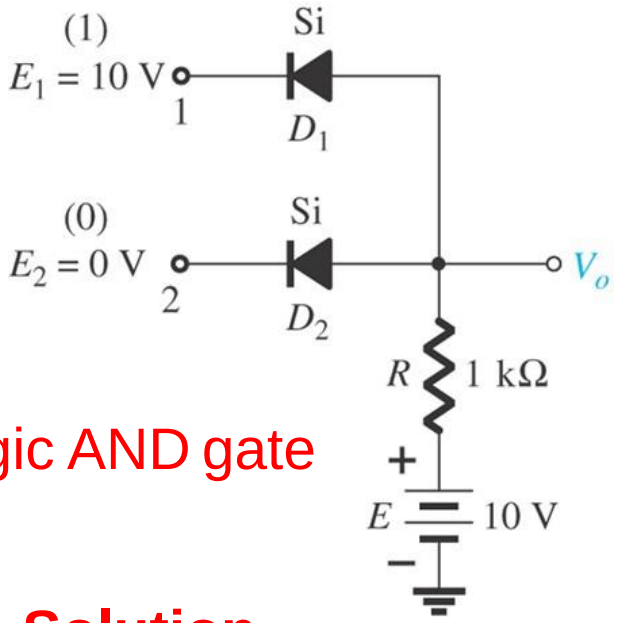
Solution



AND Gate

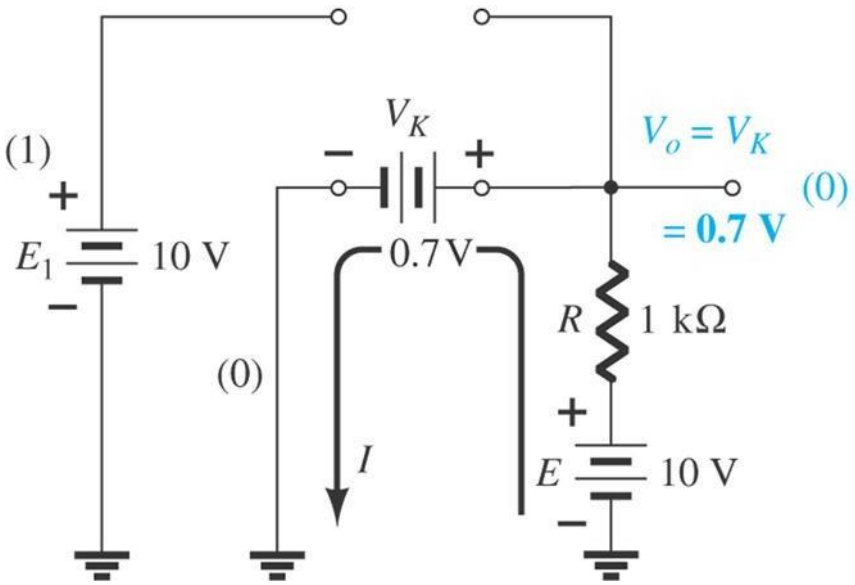
Determine V_o

Case	D1 (Si)	D2 (Si)	V_o
V_1=10 V_2=10	Reverse	Reverse	V=10 (level 1)
V_1=10 V_2=0	Reverse	Forward	V=0.7 (level 0)
V_1=0 V_2=10	Forwaed	Reverse	V=0.7 (level 0)
V_1=0 V_2=0	Forwaed	Forwaed	V=0.7 (level 0)



Logic AND gate

Solution



Next week you will deliver project 1: Part 1:

Diode-based logic gate circuits

1. Hardware circuits
2. Simulation files of all circuits
3. Mathematical analysis of all circuits

Now, compare between them